

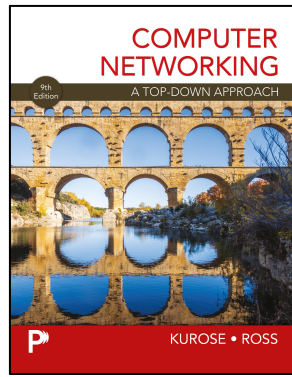
CSC 361

Computer Networks

Overview

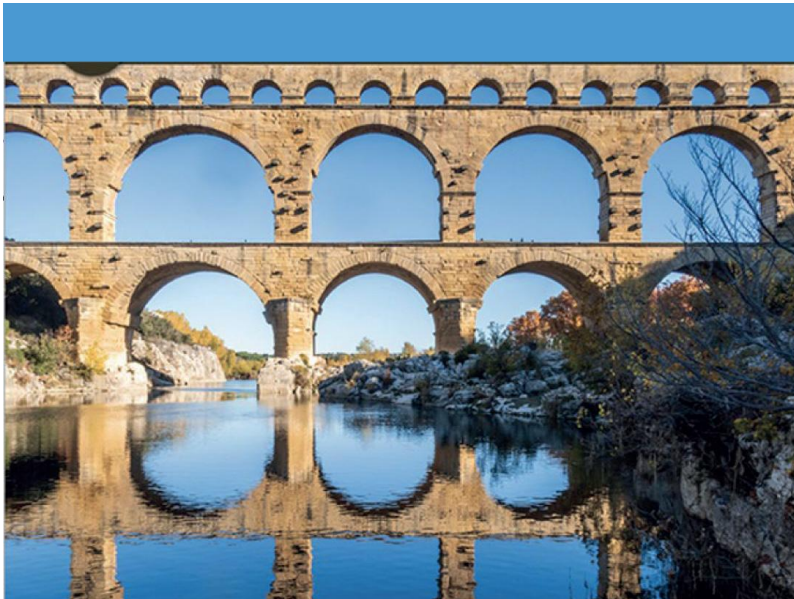
Wenjun Yang

Summer 2026



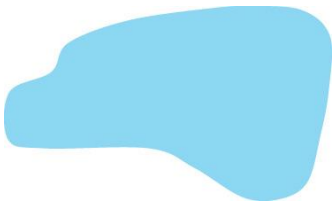
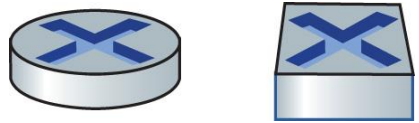
Chapter goal:

- Get “feel,” “big picture,” introduction to terminology
 - more depth, detail **later** in course



Overview/roadmap:

- What is the Internet? What is a protocol?
- **Network edge:** hosts, access network, physical media
- **Network core:** packet/circuit switching, internet structure
- **Performance:** loss, delay, throughput
- Protocol layers, service models



Billions of connected computing **devices**:

- **hosts** = end systems
- running **network apps** at Internet's "edge"

Packet switches: forward packets (chunks of data)

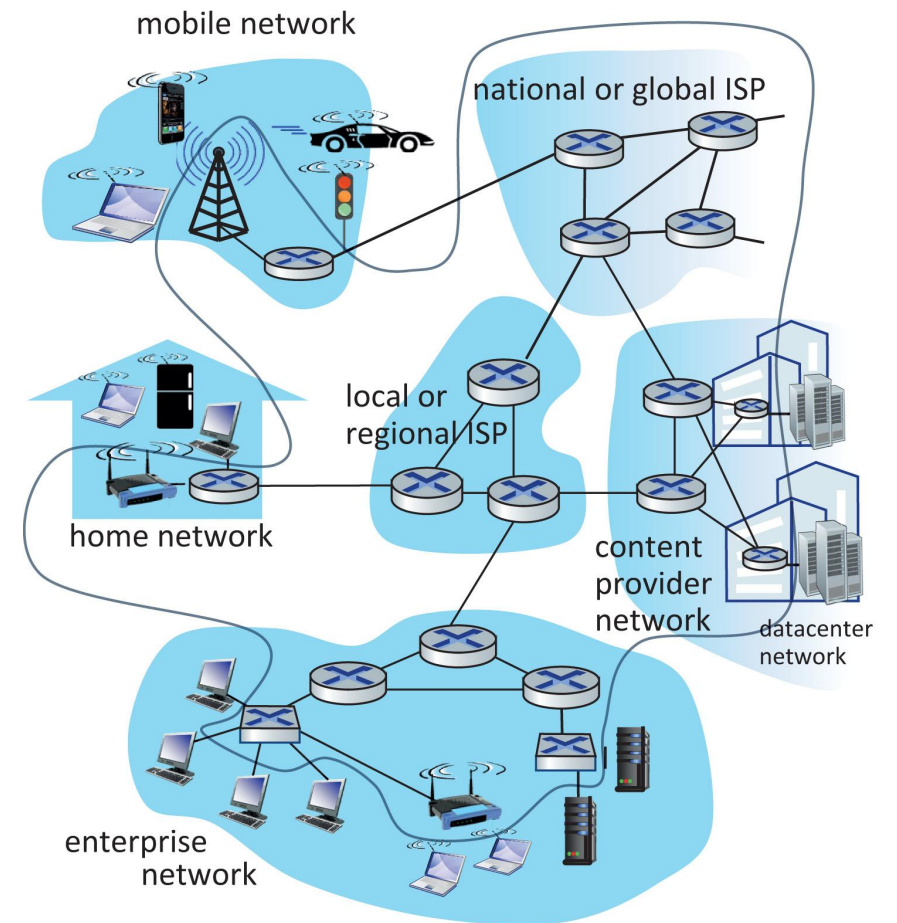
- routers, switches

Communication links

- fiber, copper, radio, satellite
- transmission rate: **bandwidth**

Networks

- collection of devices, routers, links: managed by an organization





Amazon Echo



Security Camera



Internet phones



Internet refrigerator



Gaming devices



IP picture frame



Slingbox: remote control cable TV



Pacemaker & Monitor



Web-enabled toaster +
Weather forecaster



sensorized, bed
mattress



Fitbit



AR devices



diapers



Tweet-a-watt:
monitor
energy use



bikes



cars



scooters

Others?

- **Internet: “network of networks”**

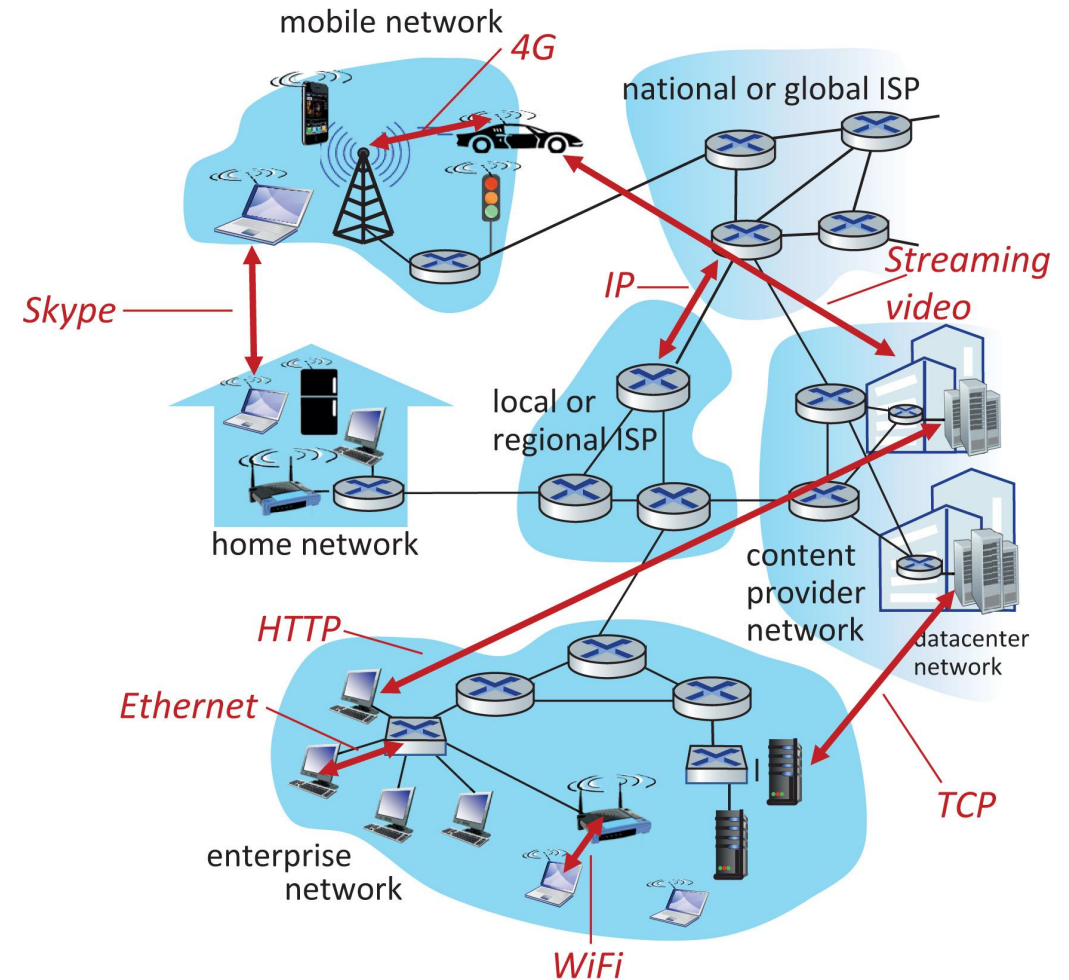
- Interconnected ISPs

- **protocols** are **everywhere**

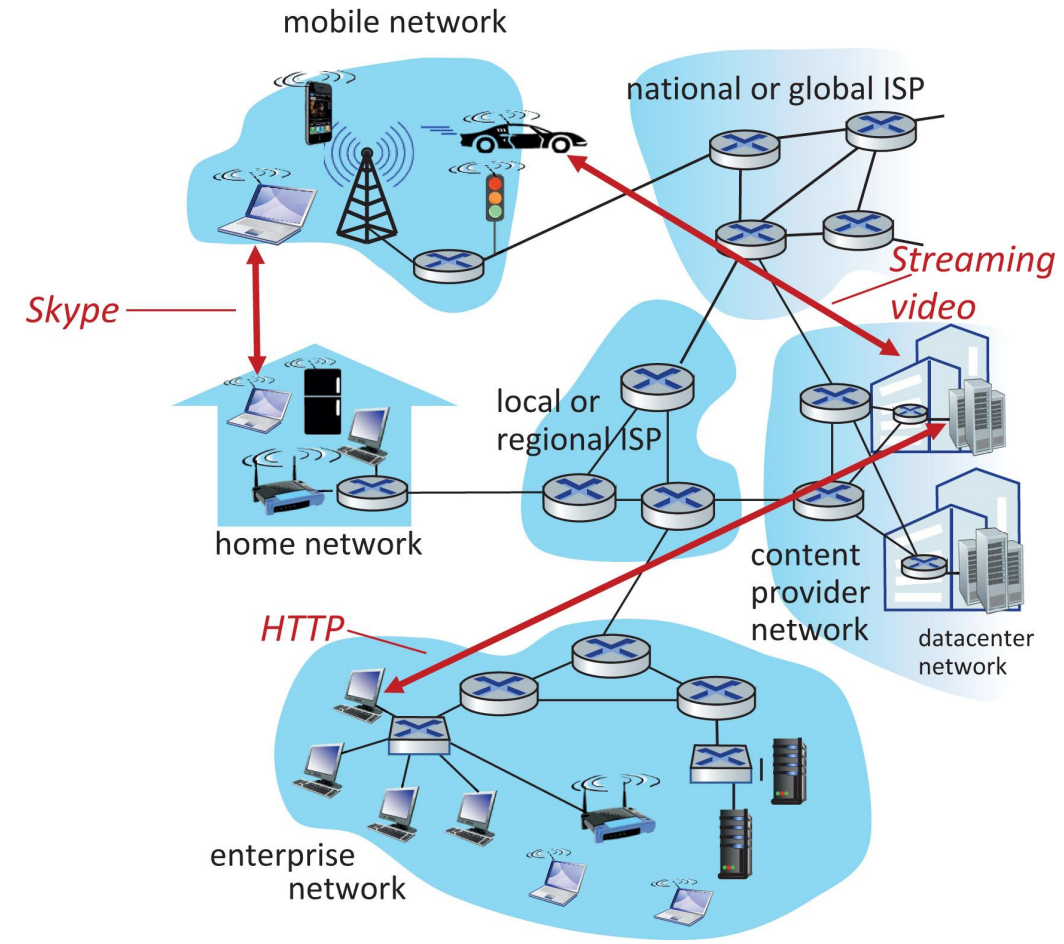
- control sending, receiving of messages
- e.g., HTTP (Web), streaming video, Zoom, TCP, IP, WiFi, 4/5G, Ethernet

- **Internet standards**

- RFC: Request for Comments
- IETF: Internet Engineering Task Force



- **Infrastructure** that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

... specific messages sent

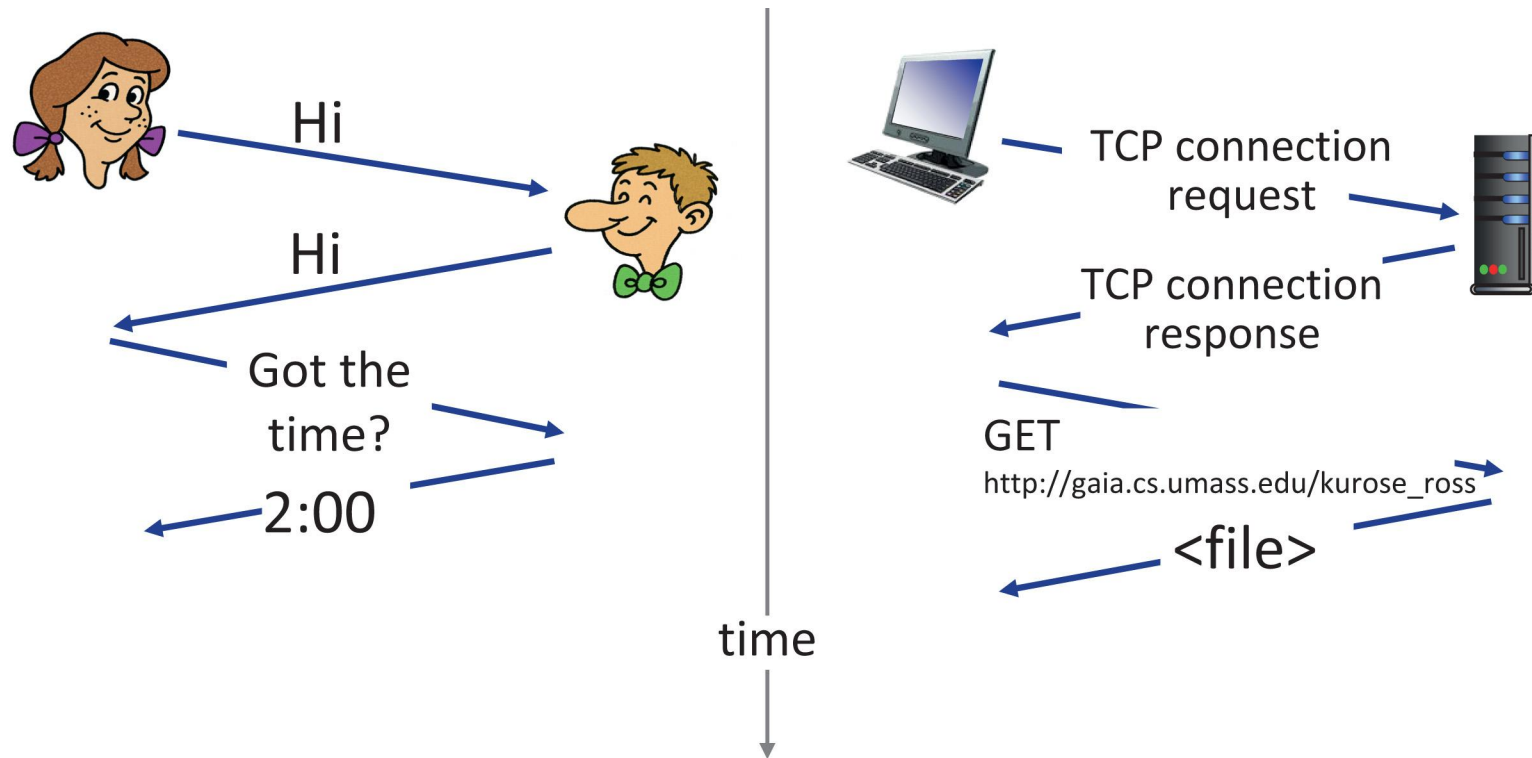
... specific actions taken when message received, or other events

Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the **format, order of messages sent and received** among network entities, and **actions taken** on message transmission, receipt

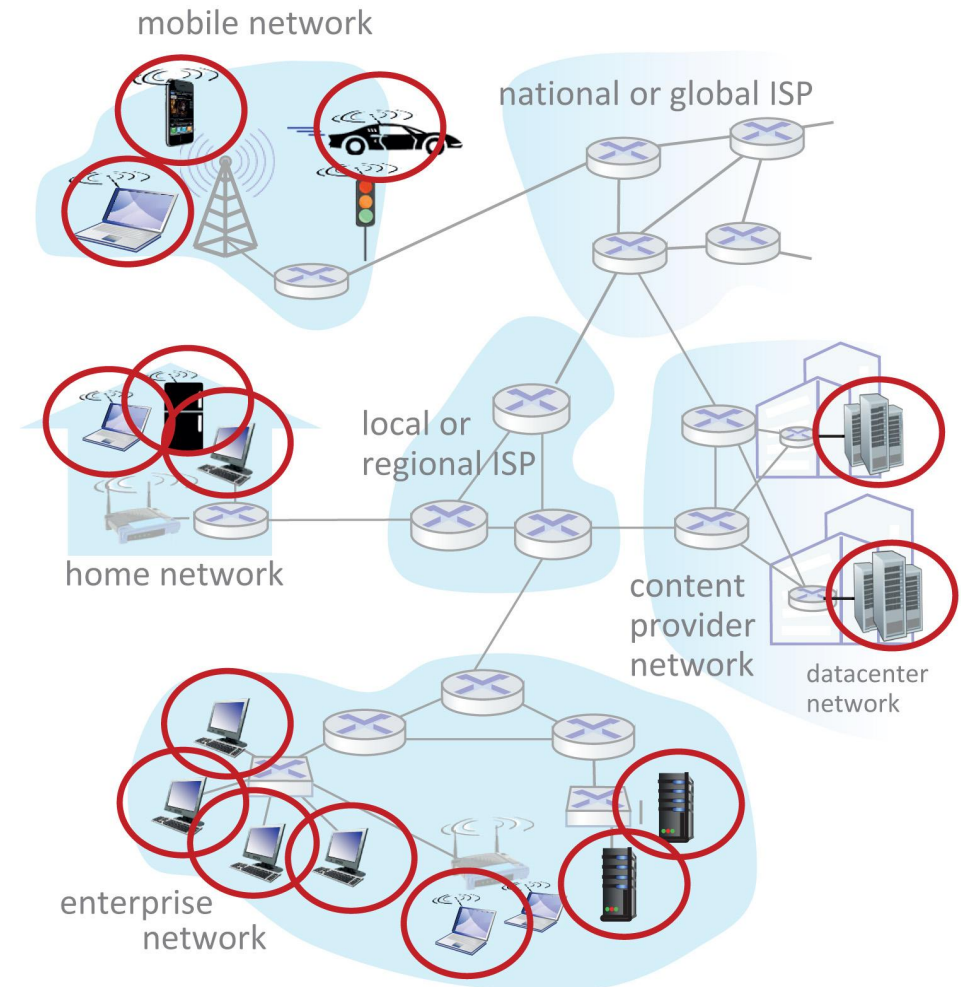
A human protocol and a computer network protocol:



- What is the Internet?
- What is a protocol?
- **Network edge:** hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History

Network edge:

- hosts: clients and servers
- servers often in data centers

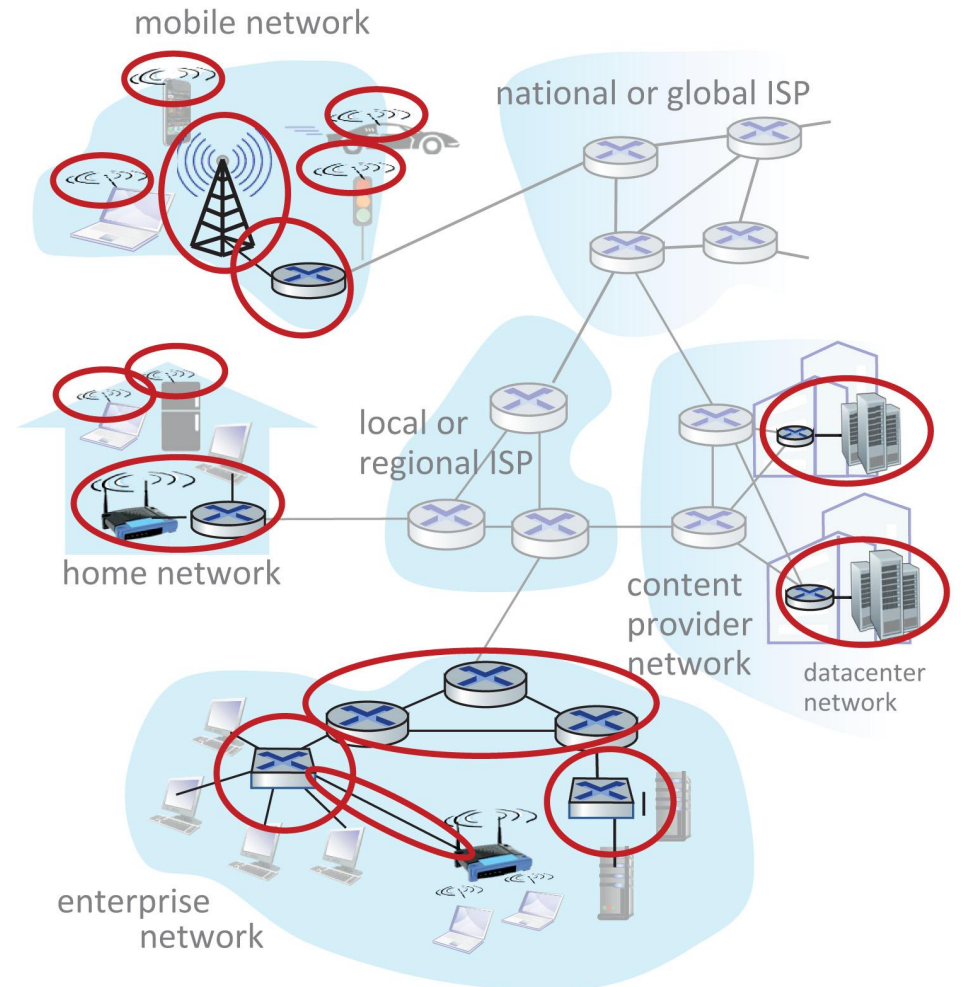


Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



Network edge:

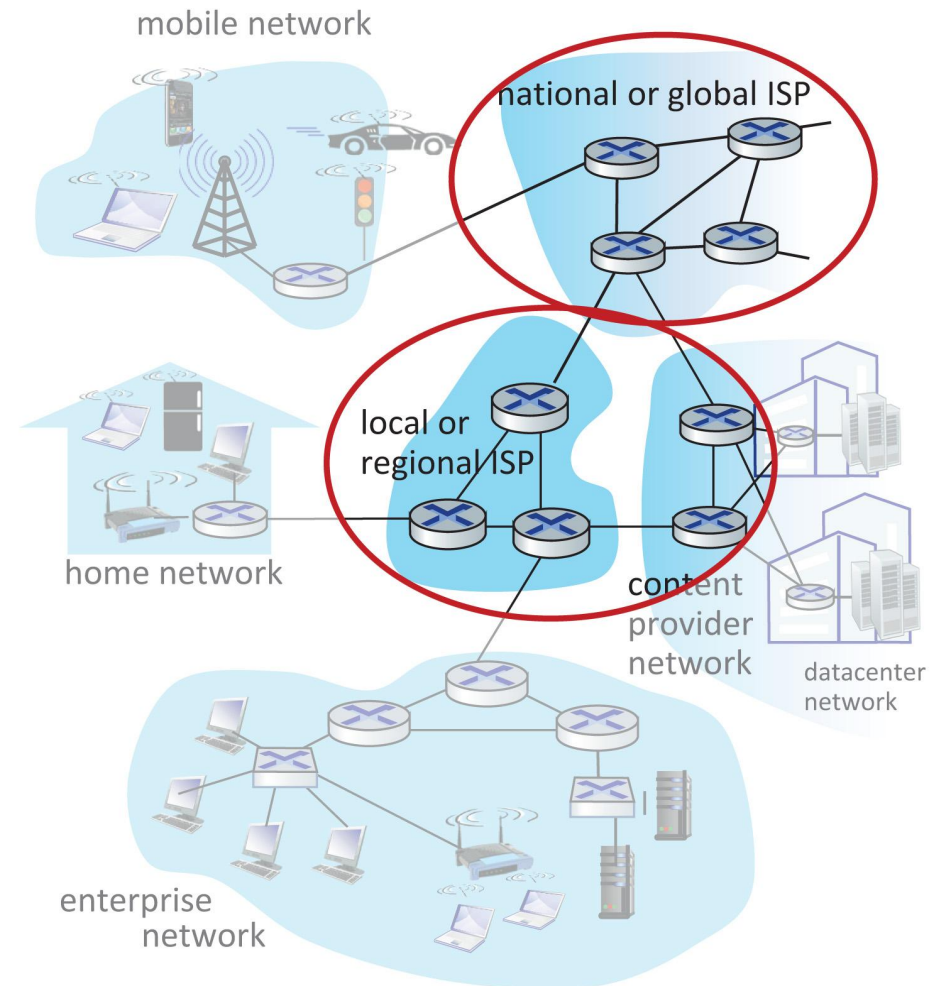
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links

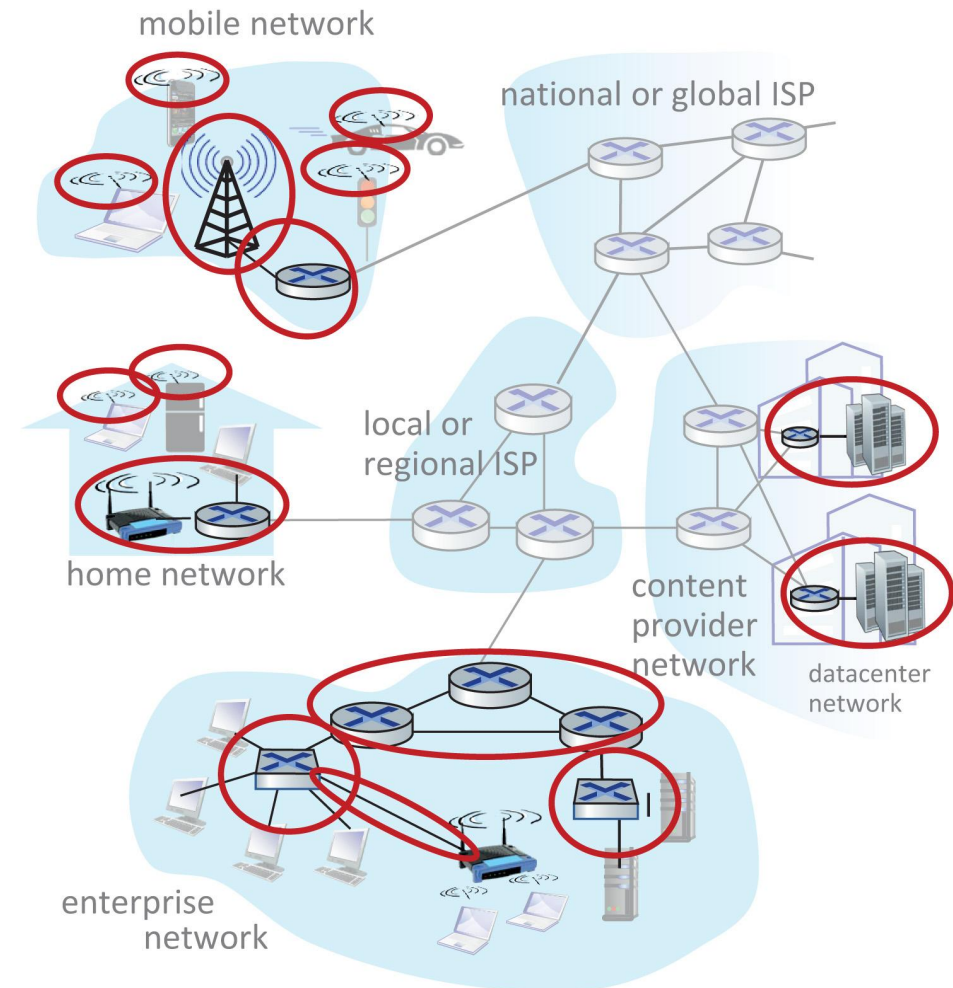
Network core:

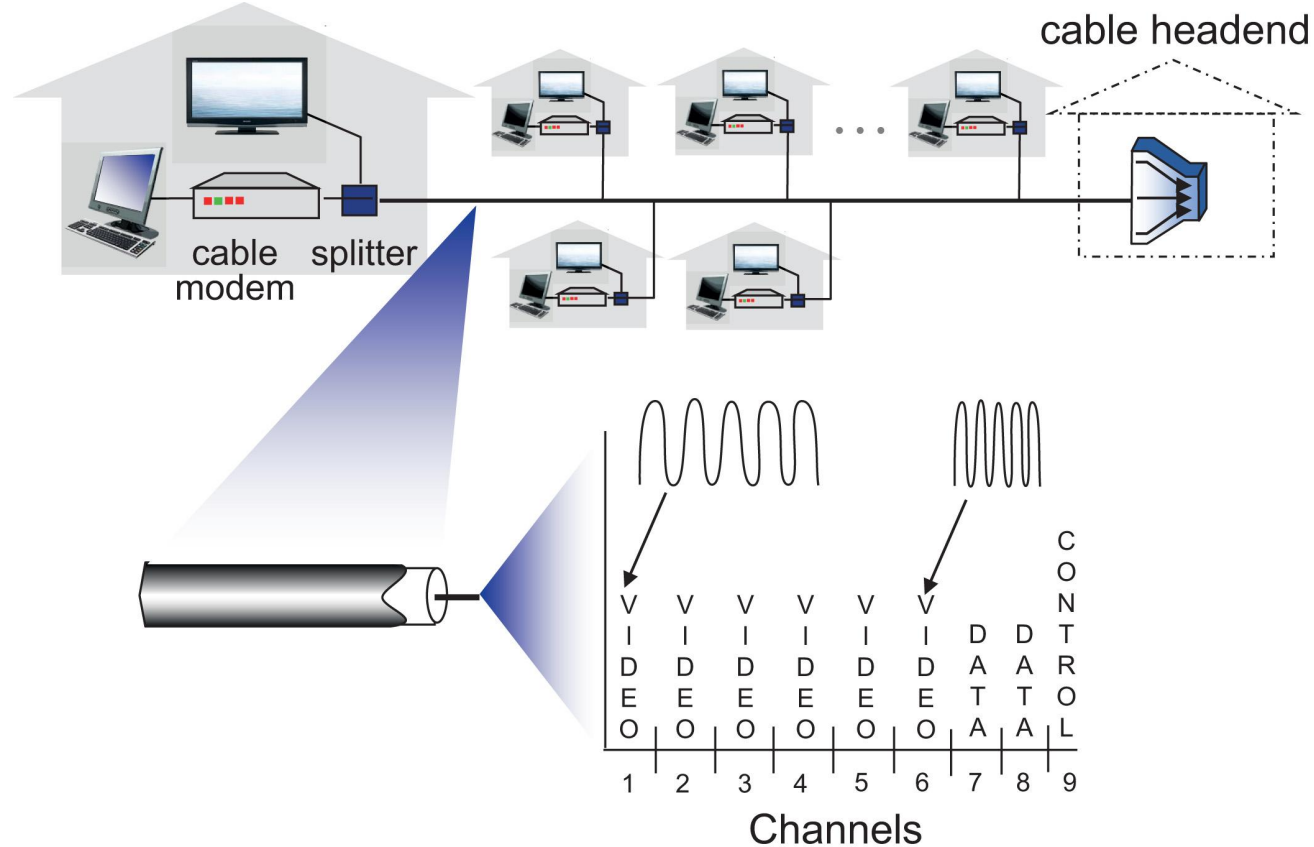
- interconnected routers
- network of networks



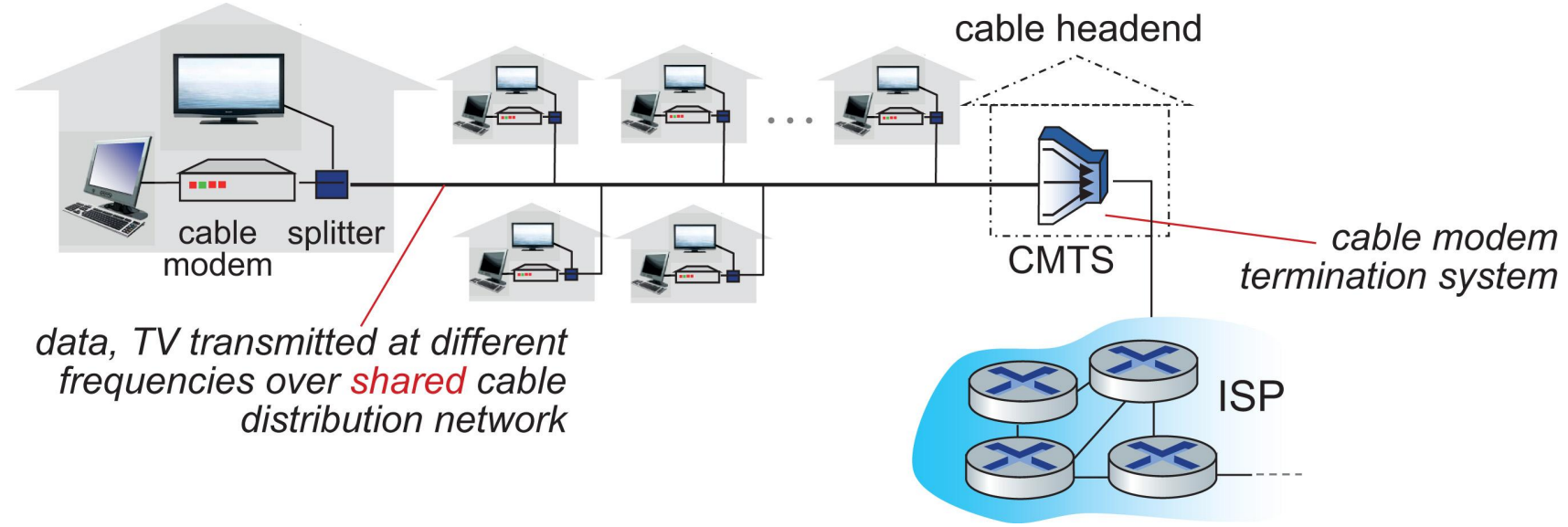
Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)





frequency division multiplexing (FDM): different channels transmitted in different frequency bands

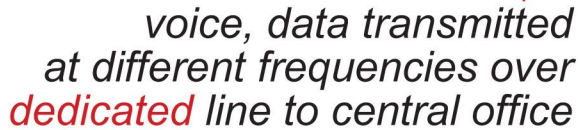


- HFC: hybrid fiber coax

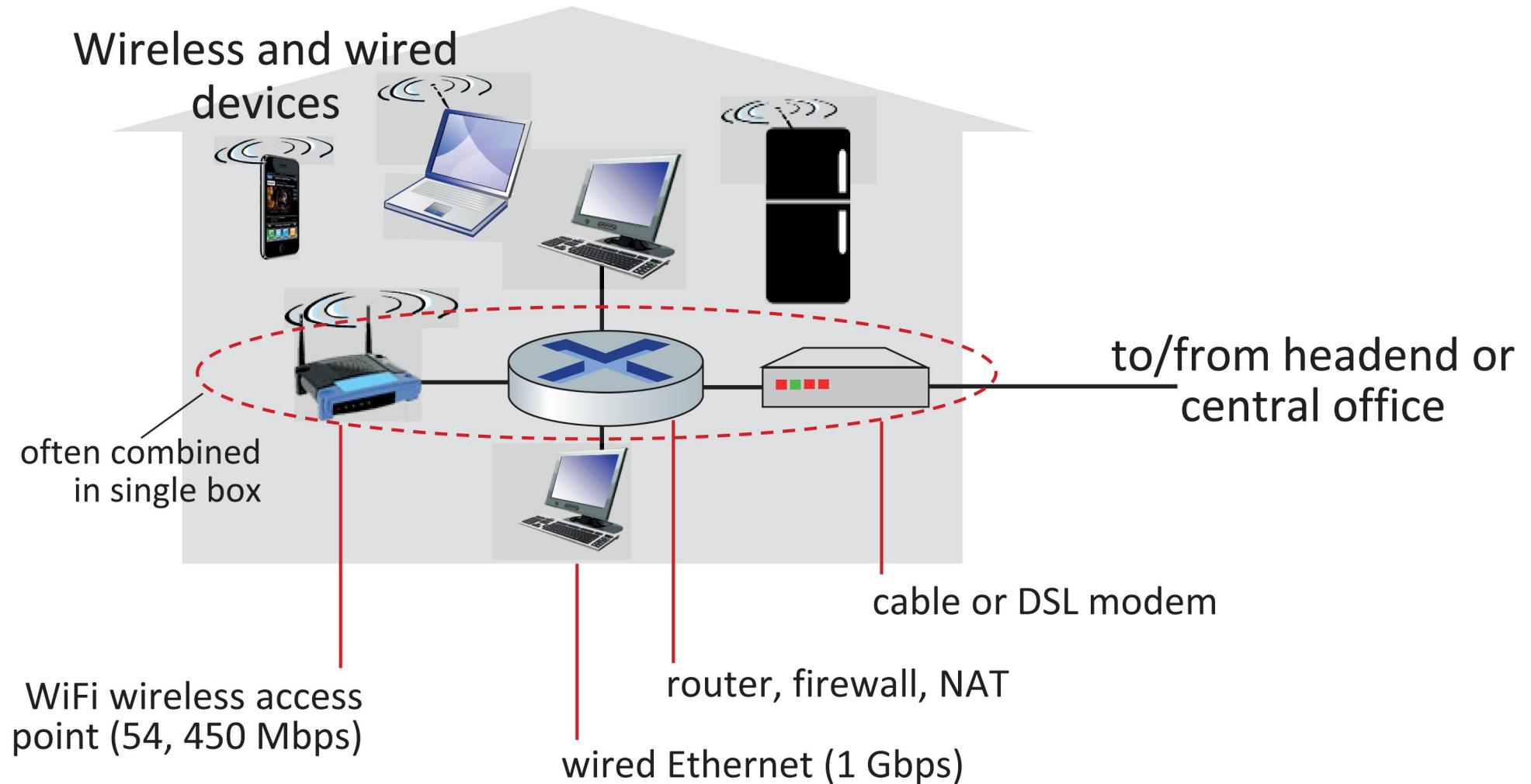
- asymmetric: up to 40 Mbps – 1.2 Gbps downstream transmission rate, 30-100 Mbps upstream transmission rate

- network of cable, fiber attaches homes to ISP router

- homes **share access network** to cable headend



- ## DSL access multiplexer



Shared **wireless** access network connects end system to router

- via base station aka “access point”

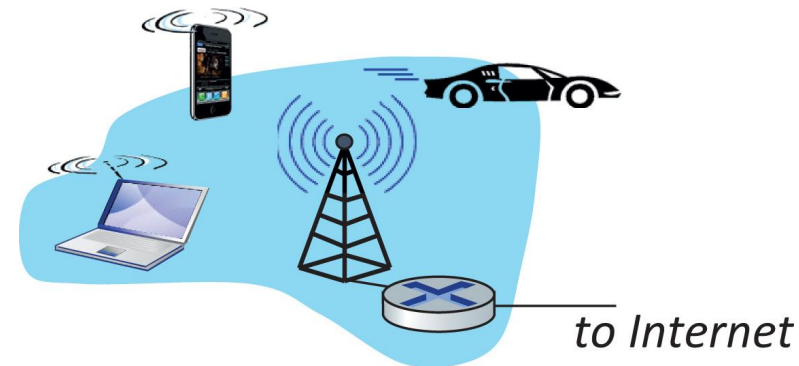
Wireless local area networks (WLANs)

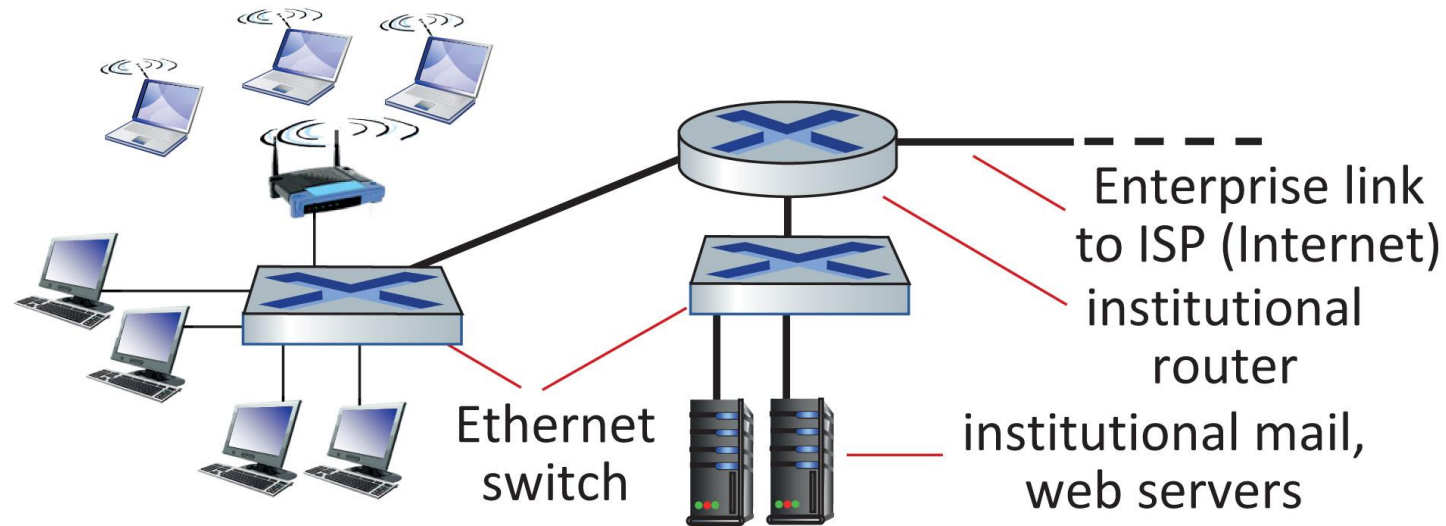
- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate



Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks

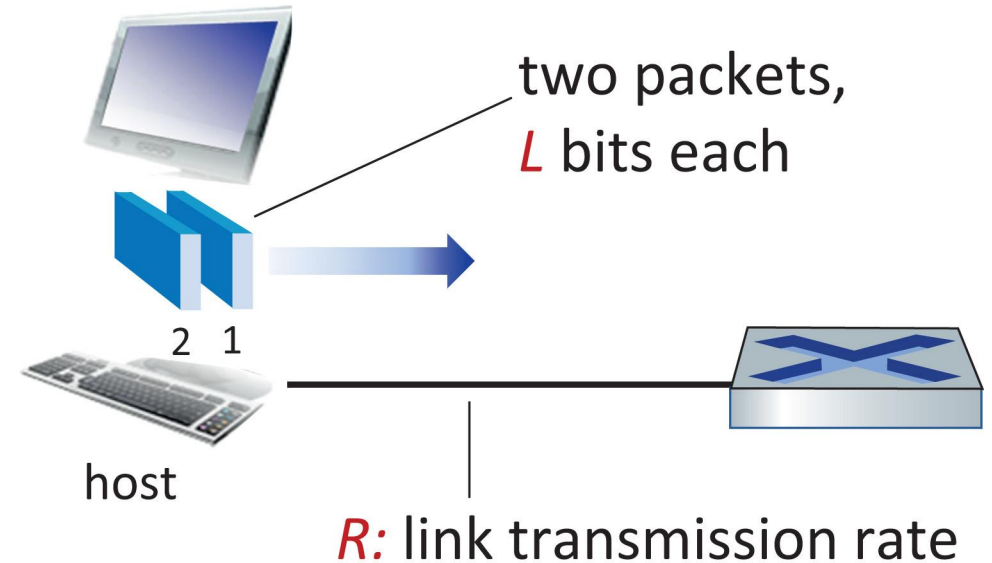




- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

host sending function:

- takes application message
- breaks into smaller chunks, known as **packets**, of length L bits
- transmits packet into access network at **transmission rate R**
 - link transmission rate, aka link **capacity**, aka link **bandwidth**

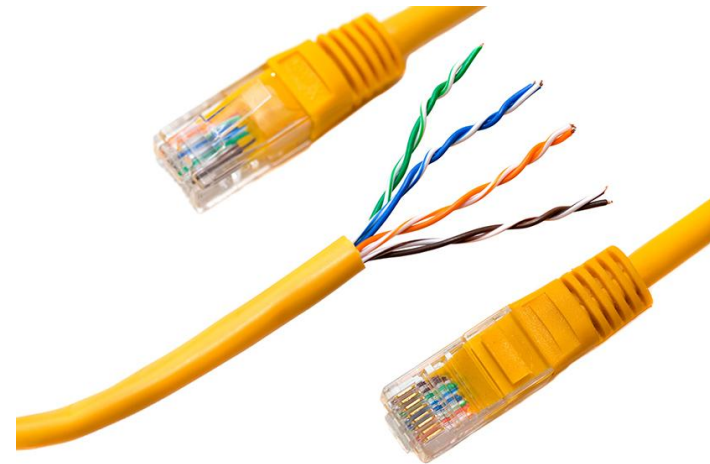


packet transmission delay = time needed to transmit L -bit packet into link = $\frac{L(\text{bits})}{R(\text{bits/sec})}$

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



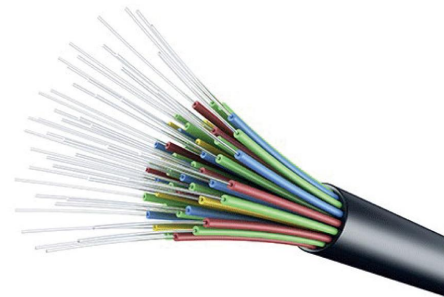
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



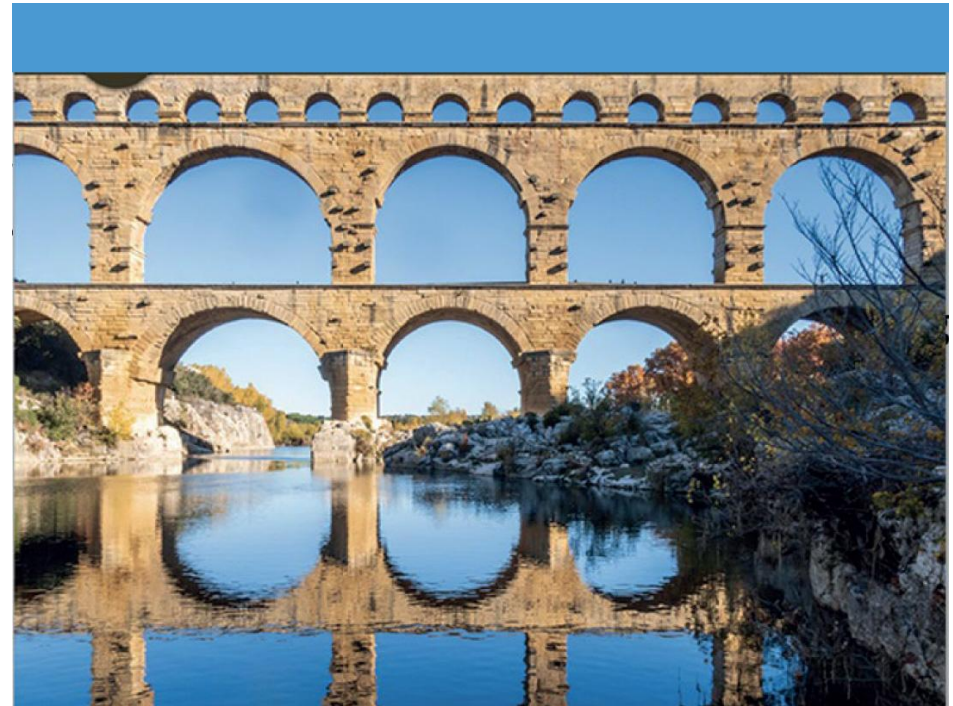
Wireless radio

- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- broadcast, “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

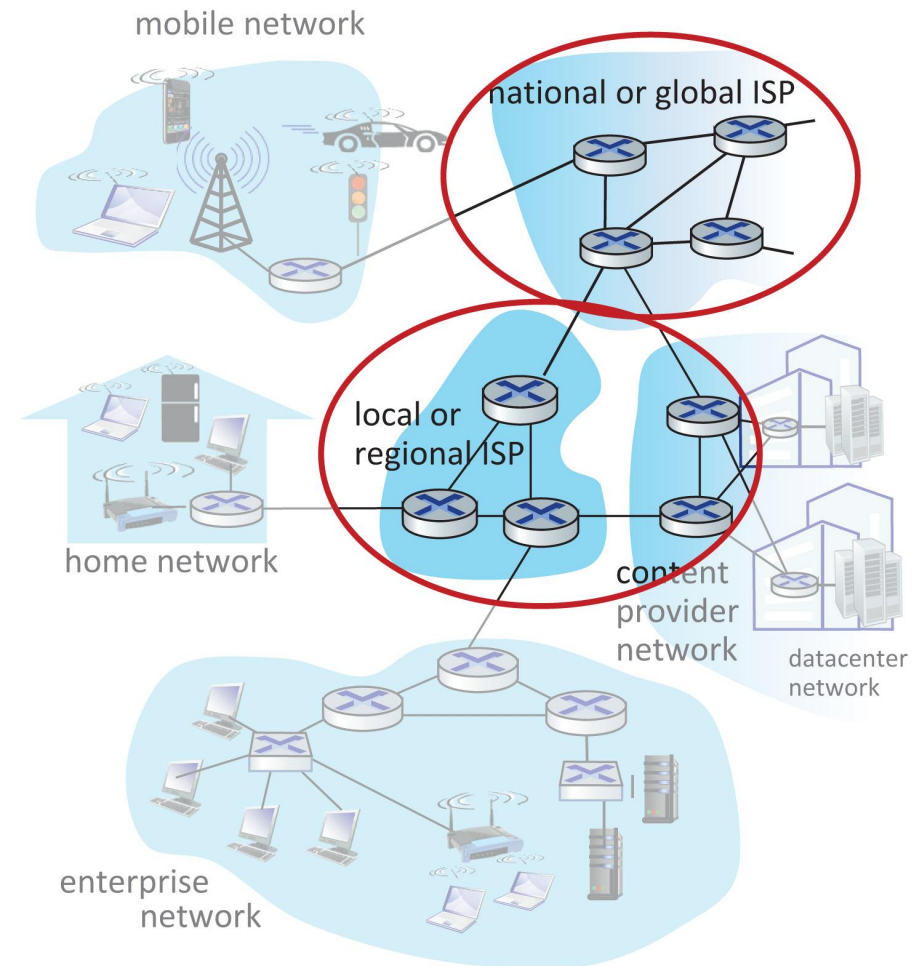
Radio link types:

- **Wireless LAN (WiFi)**
 - 10-100’s Mbps; 10’s of meters
- **wide-area** (e.g., 4G/5G cellular)
 - 100’s Mbps (4G/5G) over ~10 Km
- **Bluetooth:** cable replacement
 - short distances, limited rates
- **terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **satellite**
 - up to < 100 Mbps (Starlink) downlink
 - 270 msec end-end delay (geostationary)

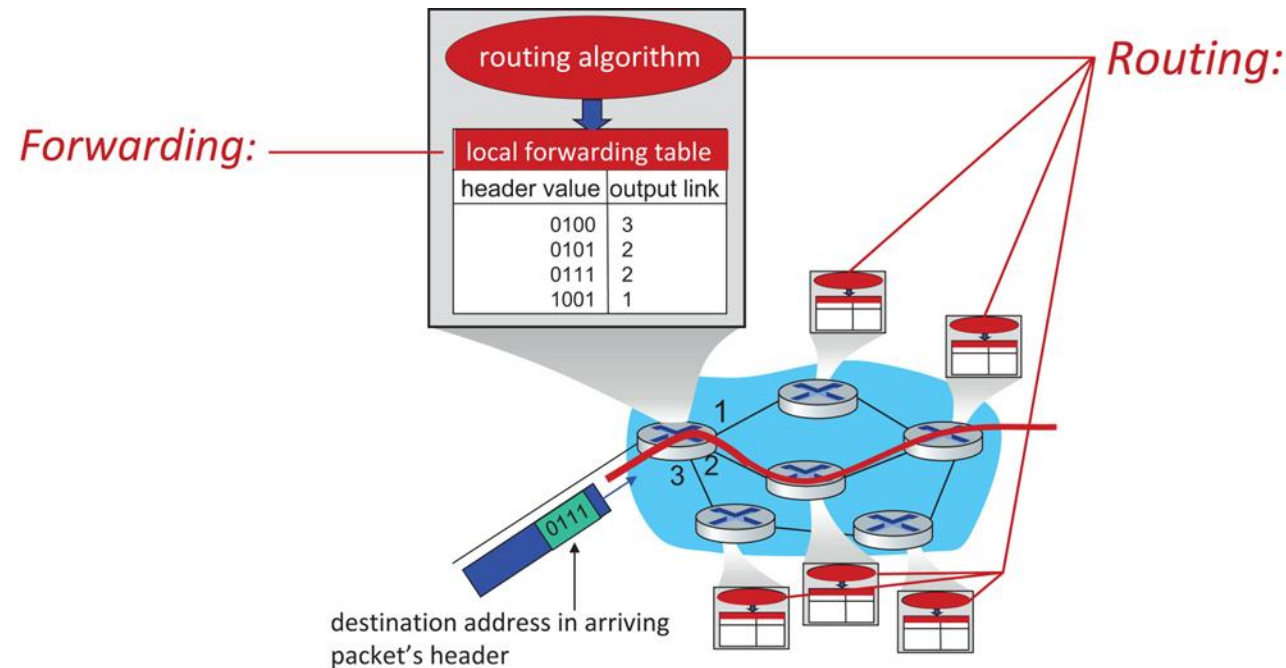
- What is the Internet?
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- mesh of interconnected routers
- **packet-switching**: hosts break application-layer messages into **packets**
 - network **forwards** packets from one router to the next, across links on path from **source to destination**



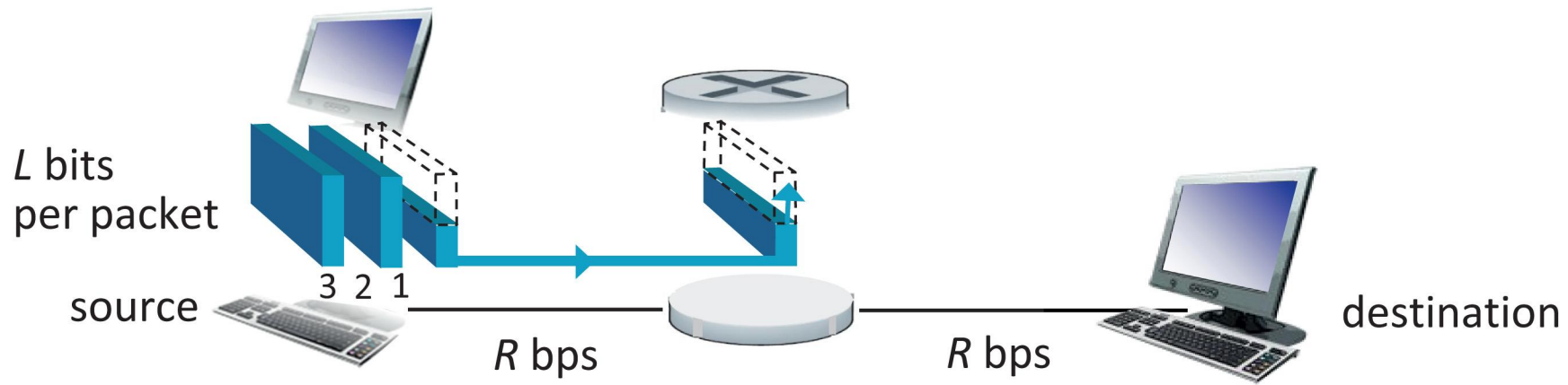
- aka “switching”
- **local** action: move arriving packets from router’s input link to appropriate router output link



- **global** action: determine source-destination paths taken by packets
- routing algorithms



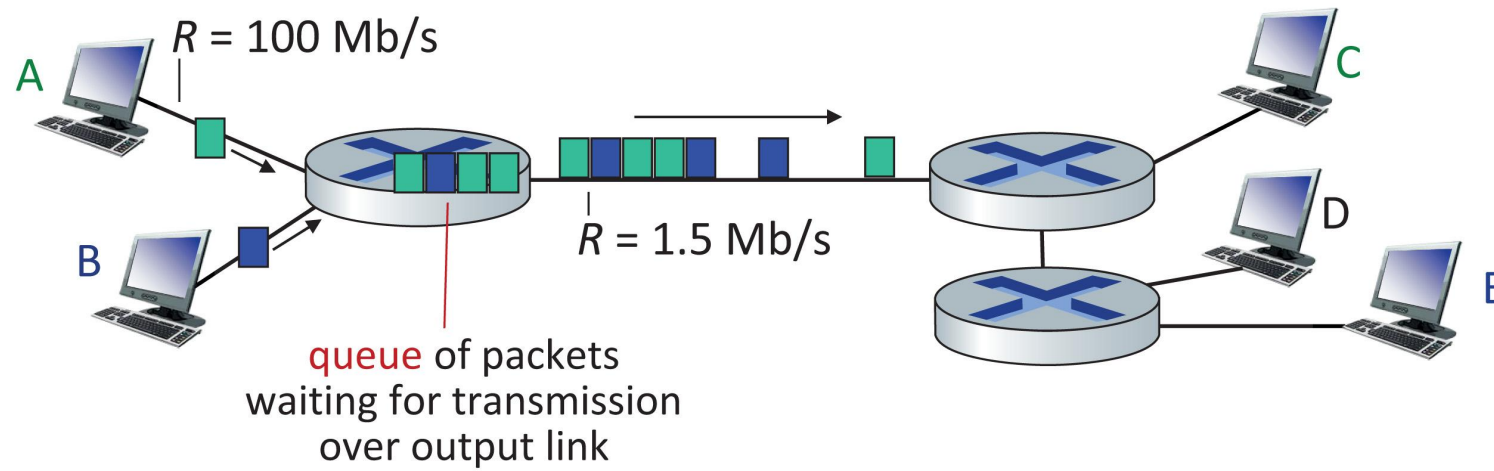




- **packet transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **store and forward:** entire packet must arrive at router before it can be transmitted on next link

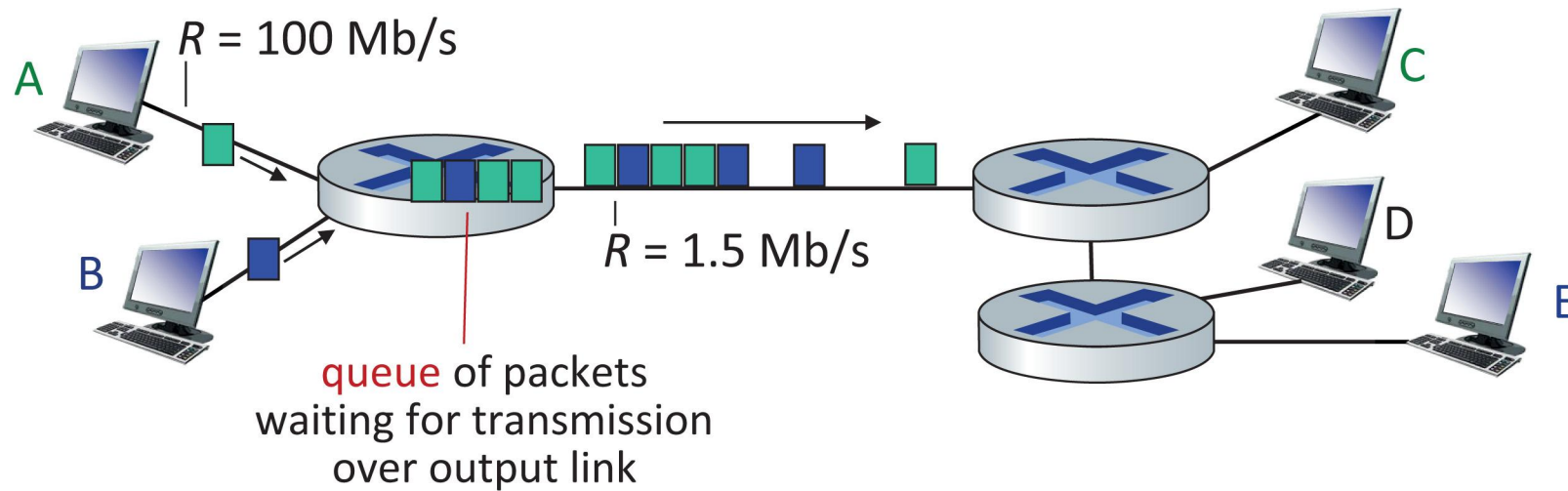
One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ M bps
- one-hop transmission delay = 0.1 msec



Queueing occurs when work arrives faster than it can be serviced:



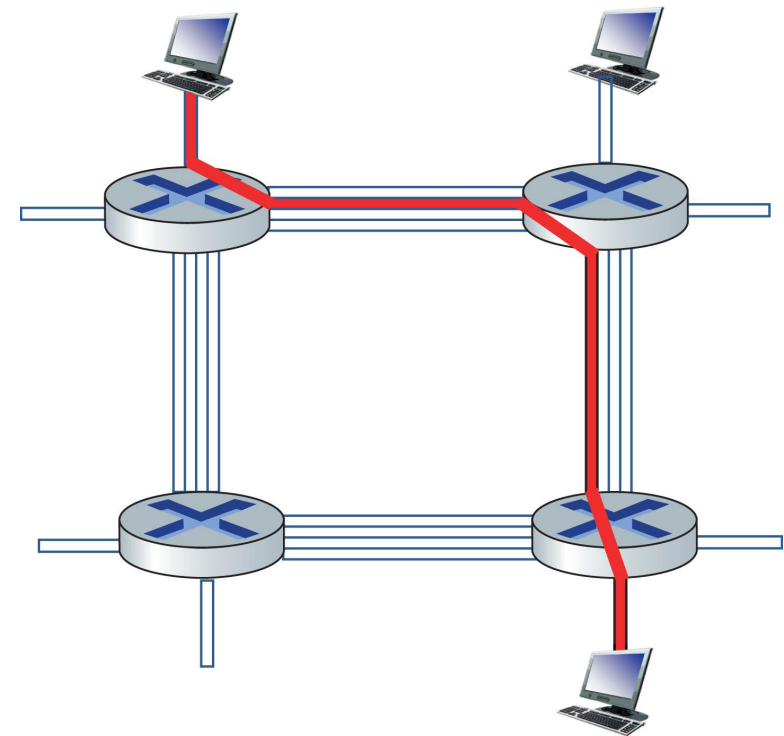


Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

end-end resources allocated to, reserved for “call”
between source and destination

- in diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks

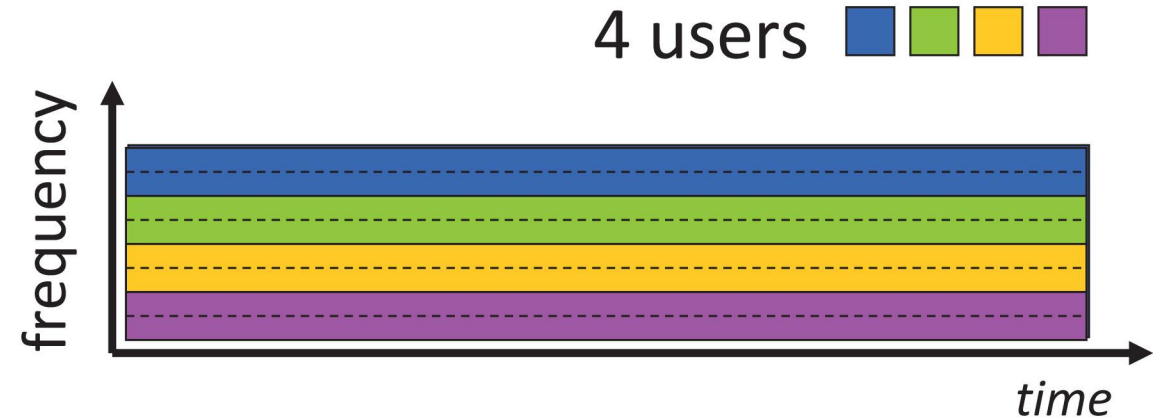


* Check out the online interactive exercises for more examples:

http://gaia.cs.umass.edu/kurose_ross/interactive

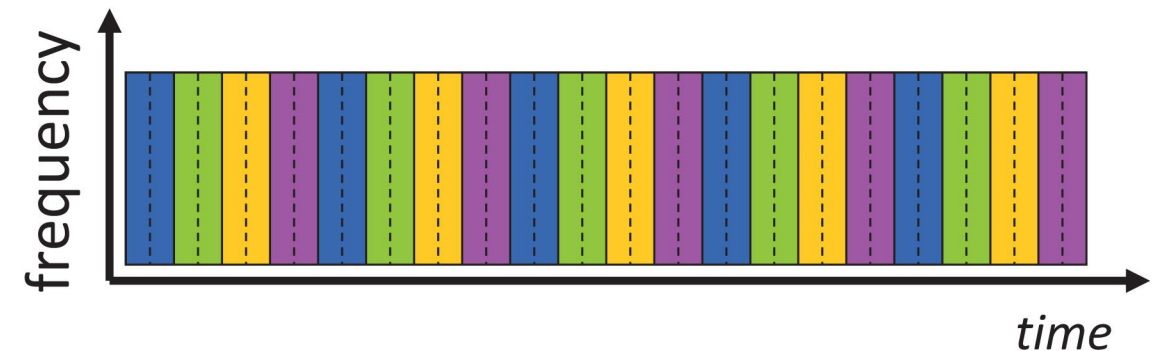
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



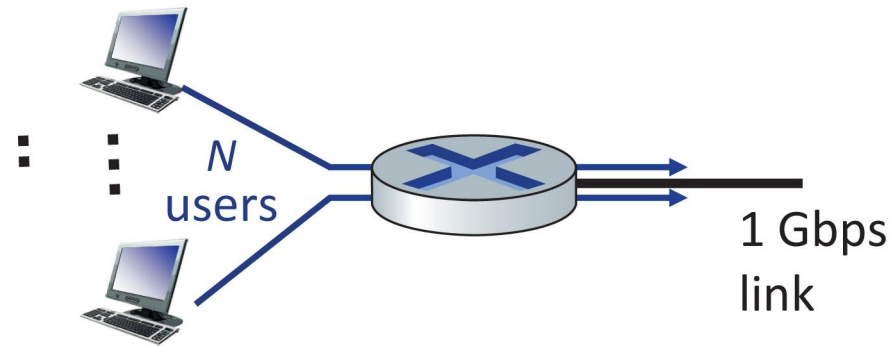
Time Division Multiplexing (TDM)

- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



example:

- 1 Gb/s link
- each user:
 - 100 Mb/s when “active”
 - active 10% of time



Q: how many users can use this network under circuit-switching and packet switching?

- **circuit-switching:** 10 users
- **packet switching:** with 35 users, probability > 10 active at same time is less than .0004 *

Q: how did we get value 0.0004?

A: HW problem (for those with course in probability only)

* Check out the online interactive exercises for more examples:

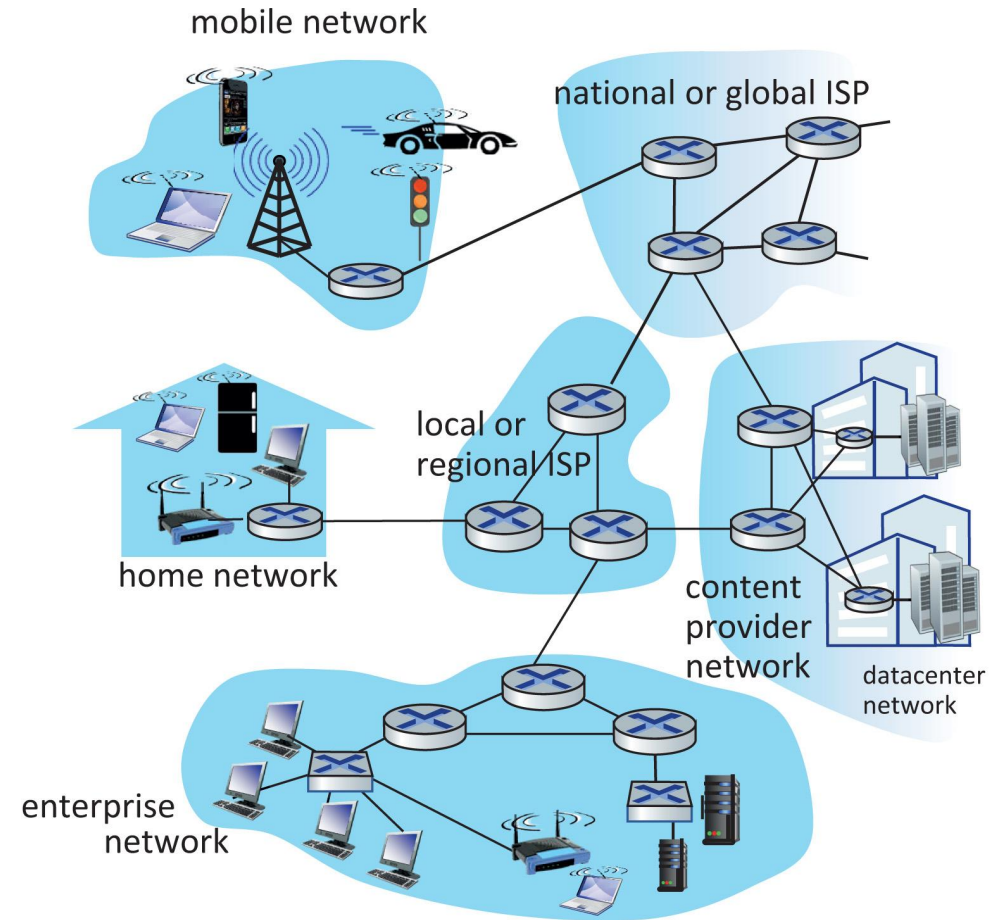
http://gaia.cs.umass.edu/kurose_ross/interactive

Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior with packet-switching?**
 - “It’s complicated.” We’ll study various techniques that try to make packet switching as “circuit-like” as possible.

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?

- hosts connect to Internet via **access** Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
 - so that **any** two hosts (**anywhere!**) can send packets to each other
- resulting network of networks is very complex
 - evolution driven by **economics**, **national policies**

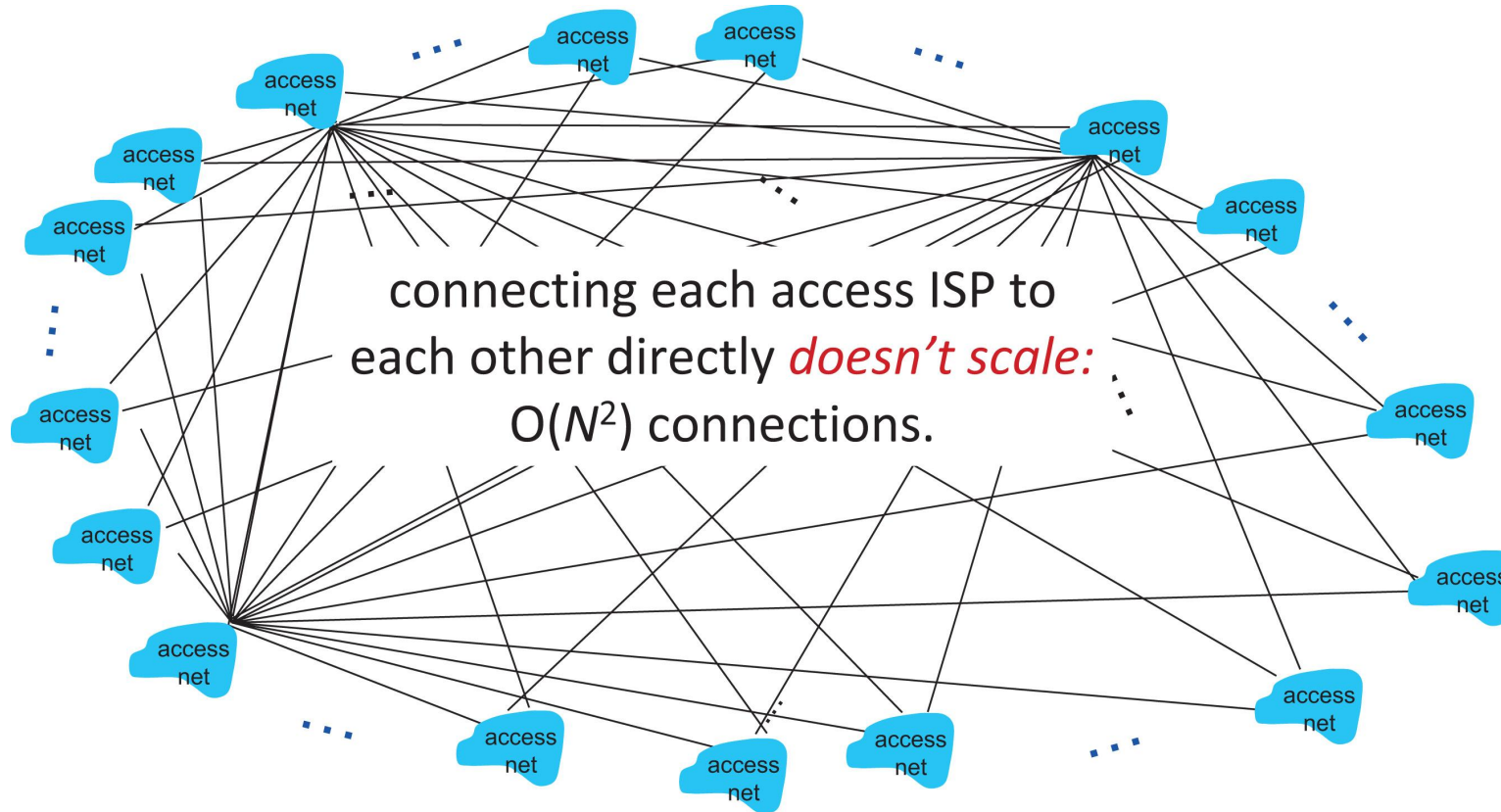


Let's take a stepwise approach to describe current Internet structure

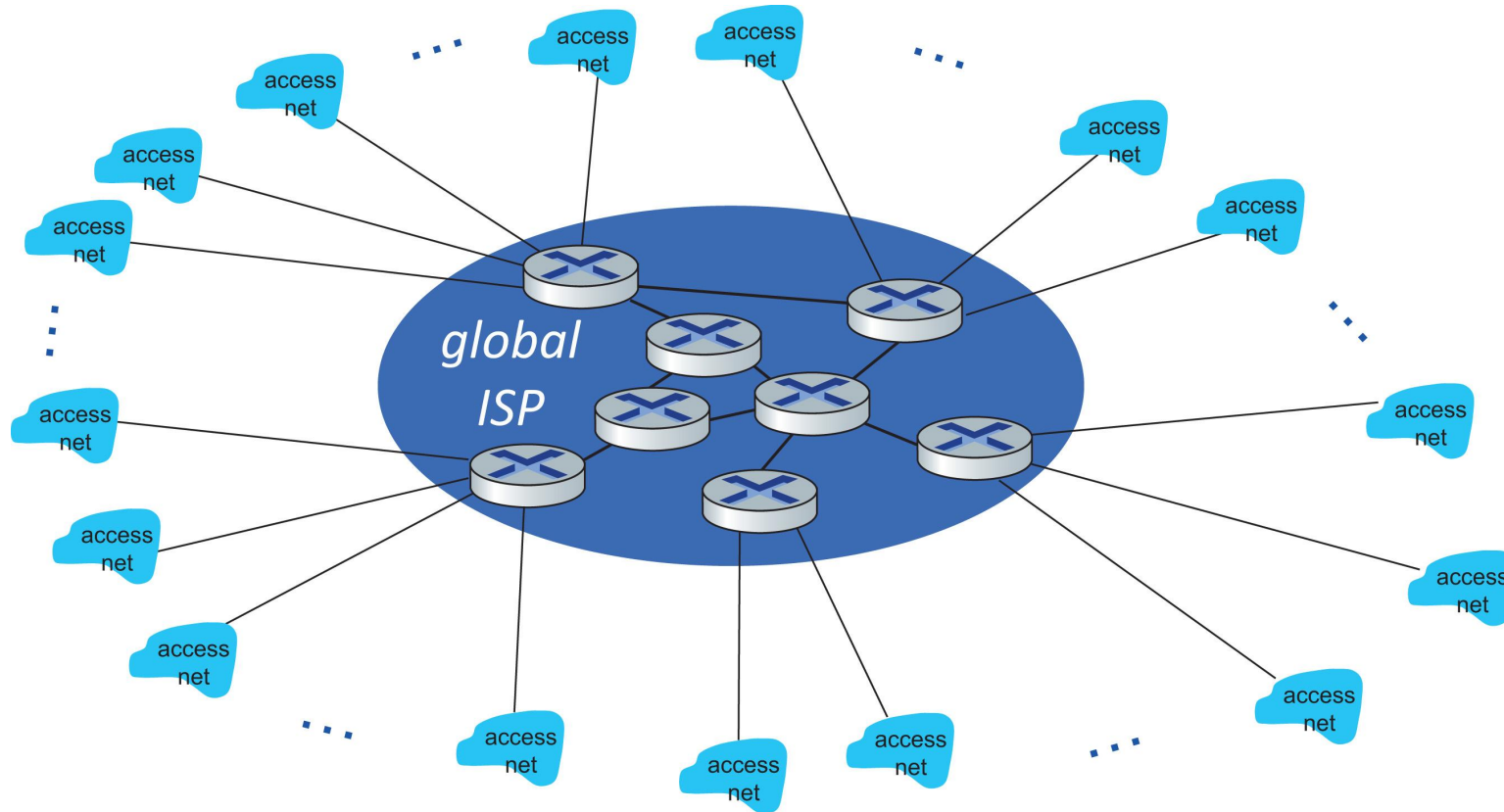
Question: given millions of access ISPs, how to connect them together?



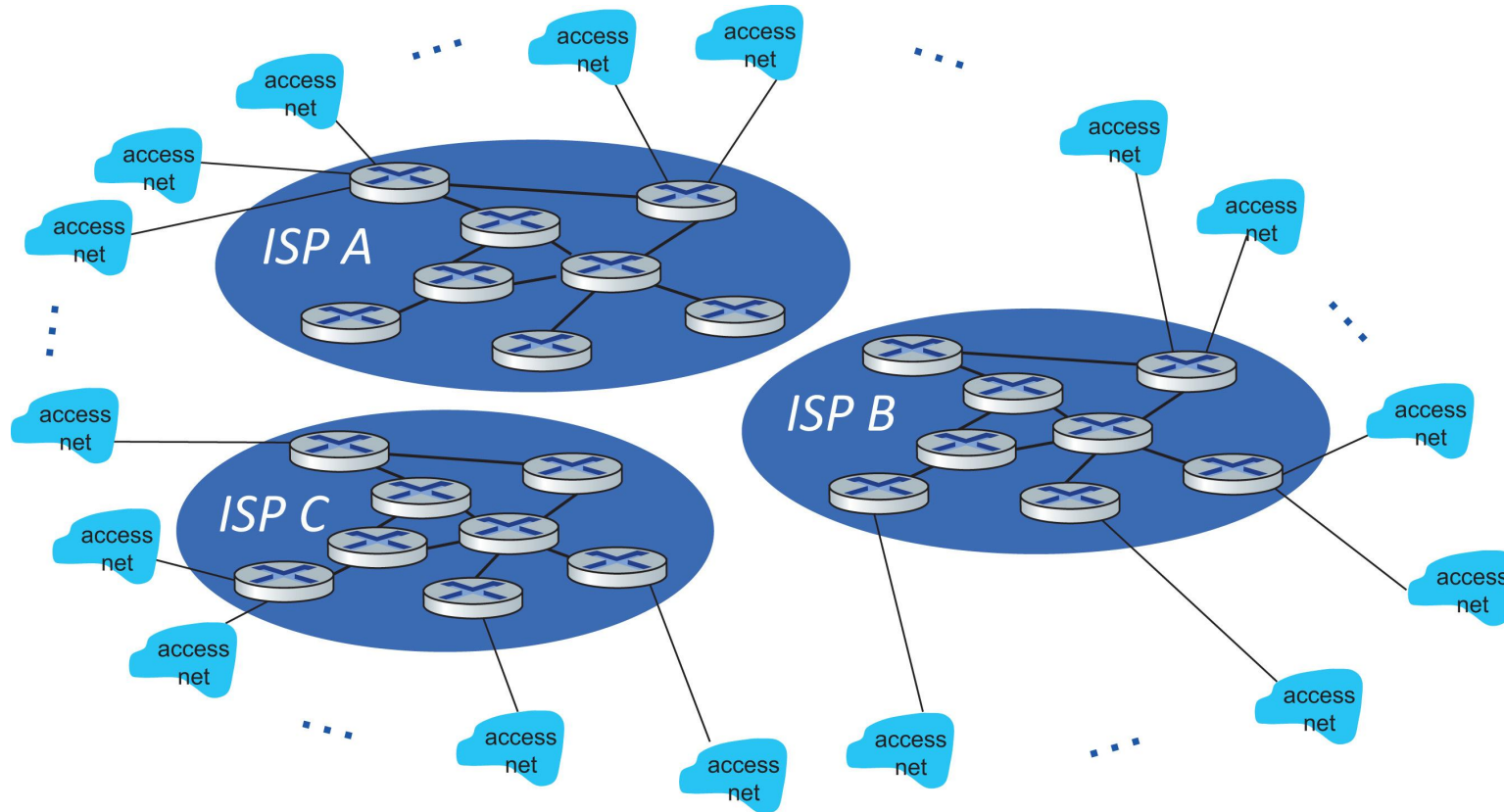
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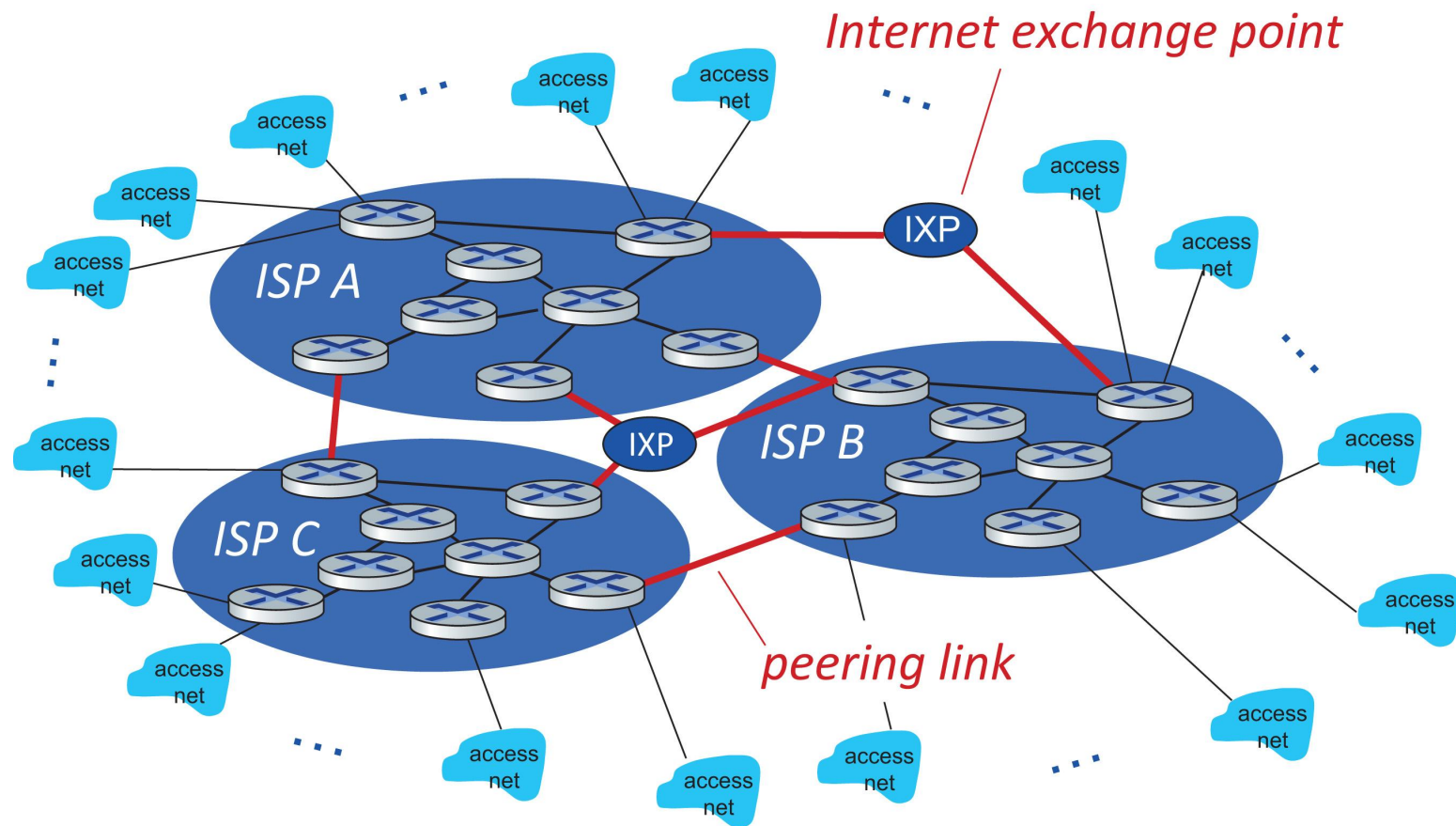
Option: connect each access ISP to one global transit ISP?
Customer and **provider** ISPs have economic agreement.



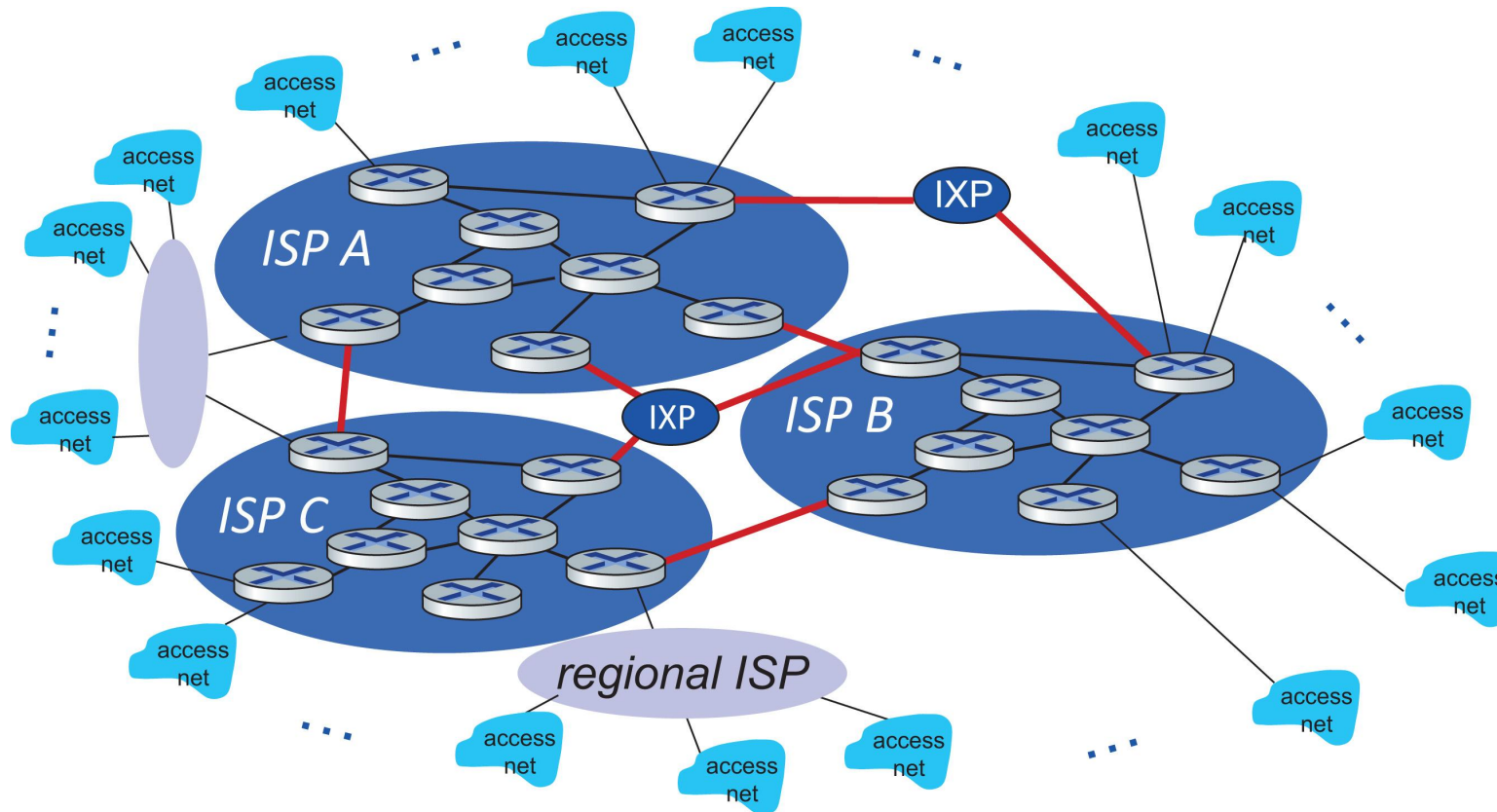
But if one global ISP is viable business, there will be competitors



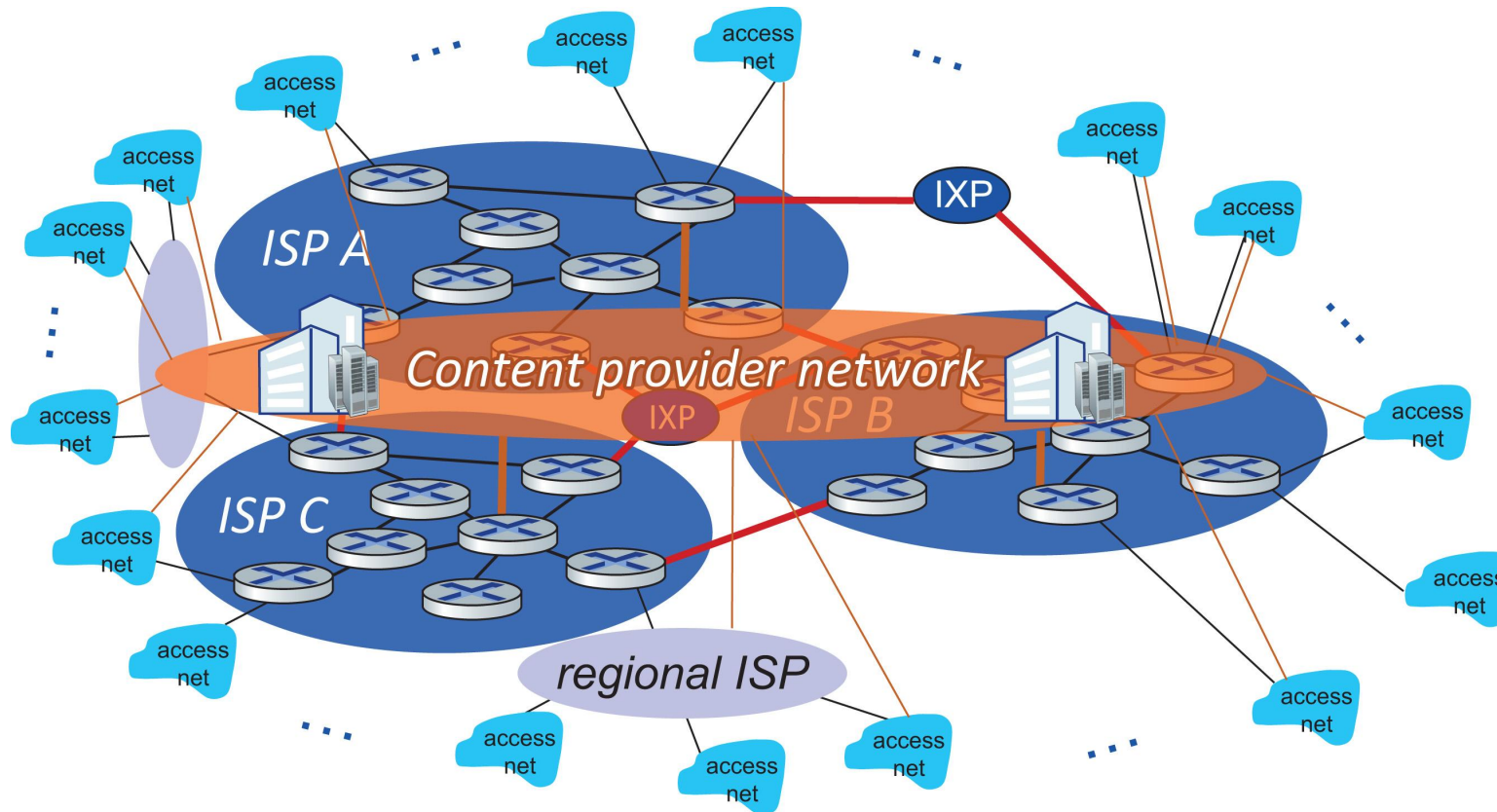
But if one global ISP is viable business, there will be competitors who will want to be connected

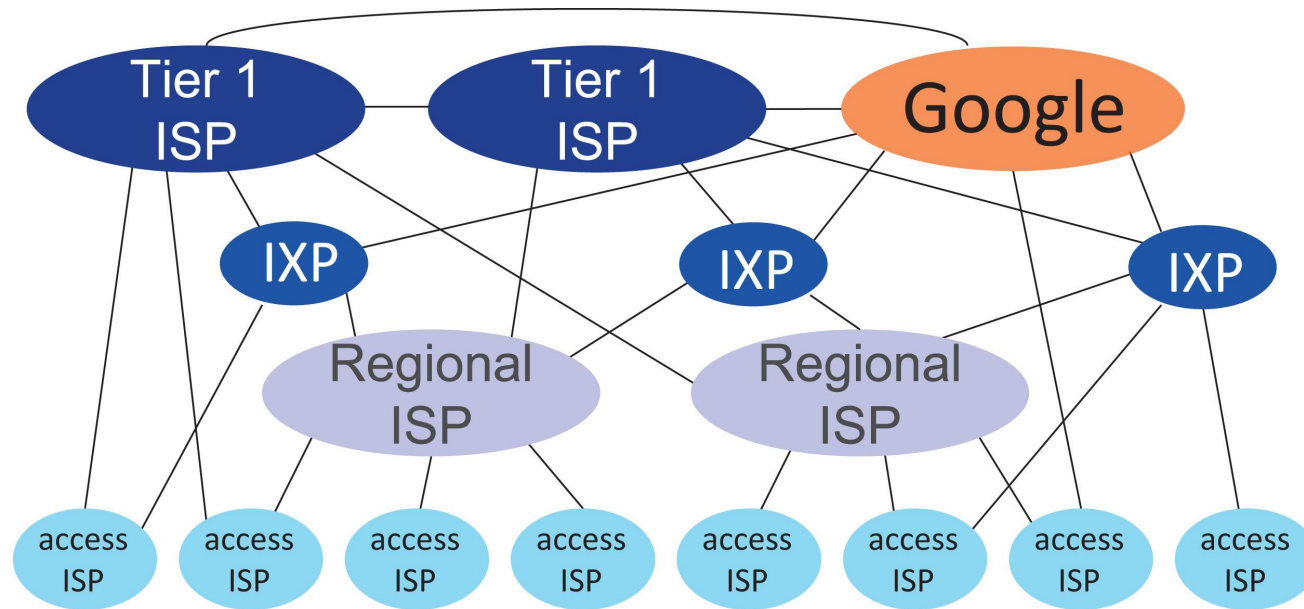


... and regional networks may arise to connect access nets to ISPs



... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users





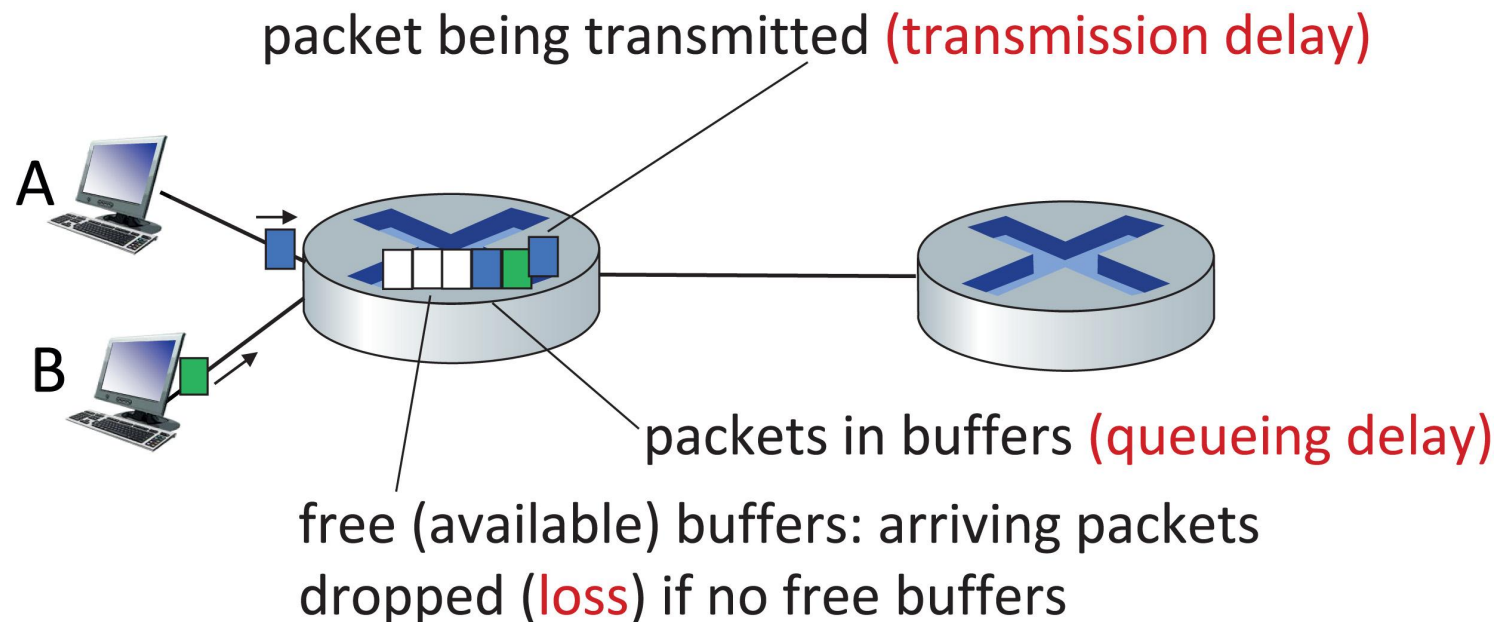
At “center”: small # of well-connected large networks

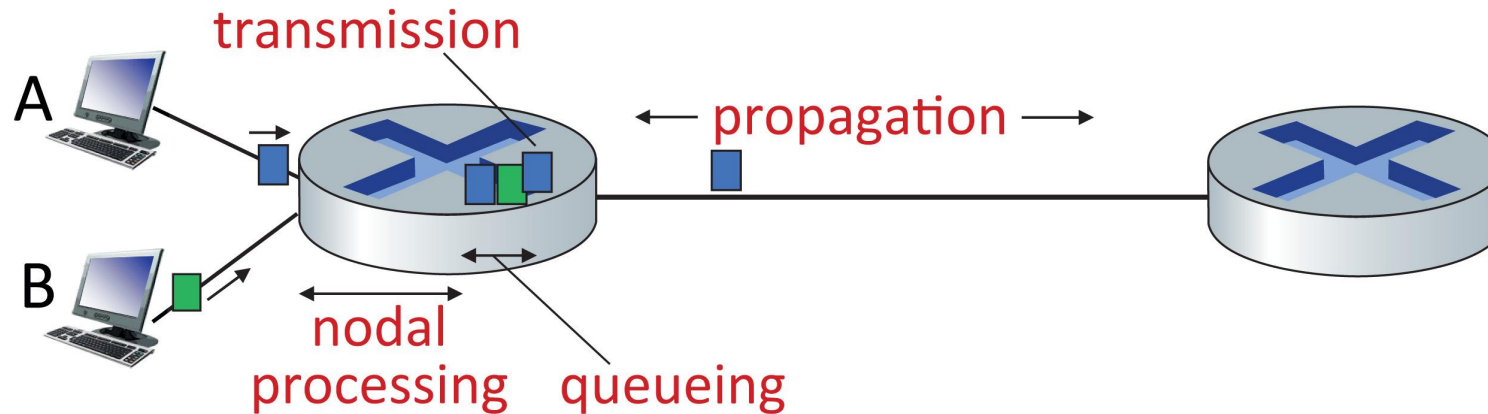
- “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

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- packets **queue** in router buffers, waiting for turn for transmission
 - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet **loss** occurs when memory to hold queued packets fills up





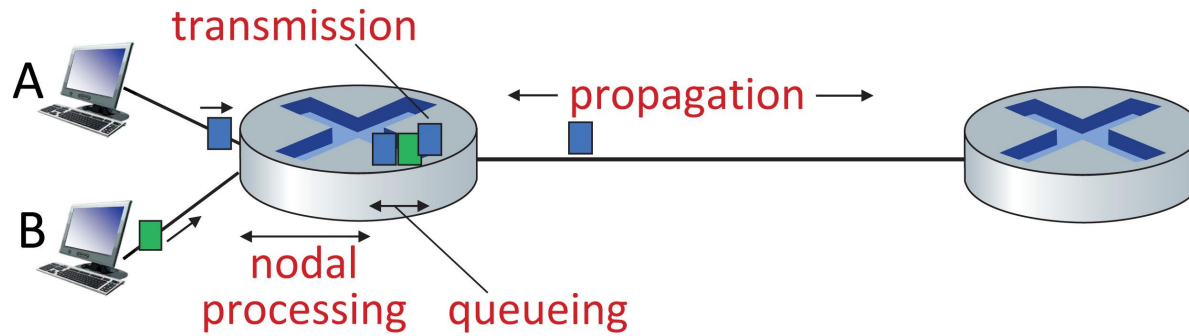
$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < microsecs

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link transmission rate (bps)

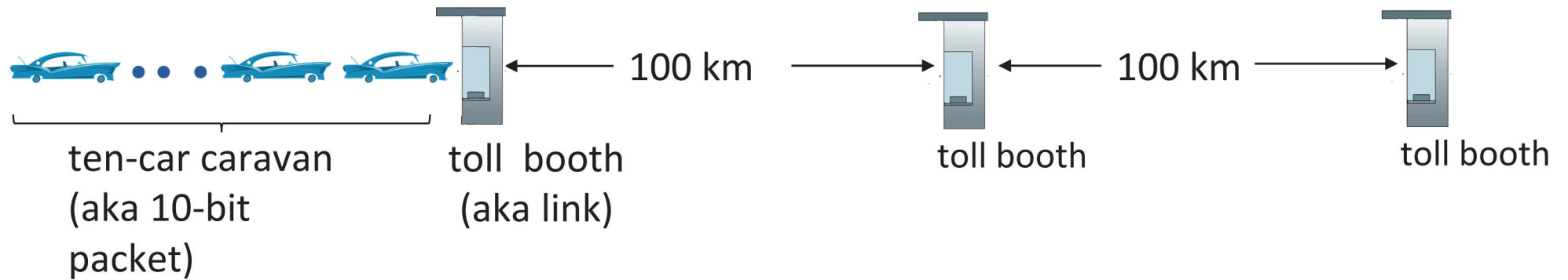
d_{proc} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8 \text{ m/sec}$)

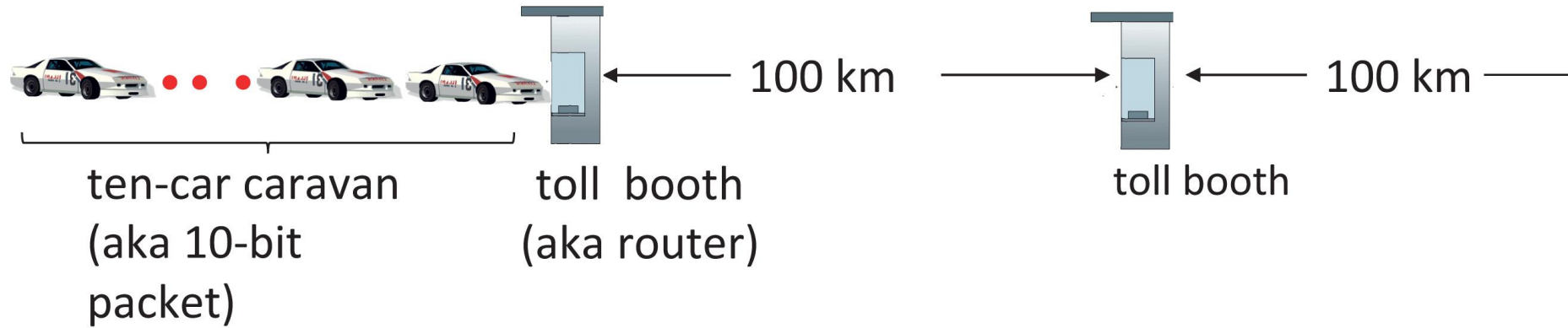
• $d_{\text{trans}} = L/R$

• $d_{\text{proc}} = d/s$

d_{trans} and d_{prop}
very different



- car ~ bit; caravan ~ packet;
toll service ~ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- “propagate” at 100 km/hr
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway
 $= 12 \times 10 = 120 \text{ sec}$
- time for last car to propagate from 1st to 2nd toll booth:
 $100 \text{ km} / (100 \text{ km} / \text{hr}) = 1 \text{ hr}$
- **A: 62 minutes**



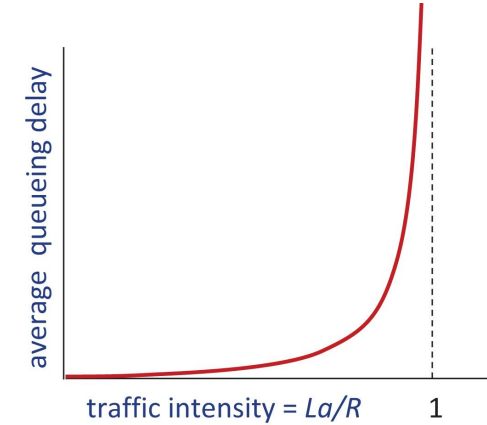
- suppose cars now “propagate” at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**

A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

- a : average packet arrival rate
- L : packet length (bits)
- R : link bandwidth (bit transmission rate)

$\frac{L \cdot a}{R}$: $\frac{\text{arrival rate of bits}}{\text{service rate of bits}}$ “traffic intensity”

- $La/R \sim 0$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay large
- $La/R > 1$: more “work” arriving is more than can be serviced – average delay infinite!

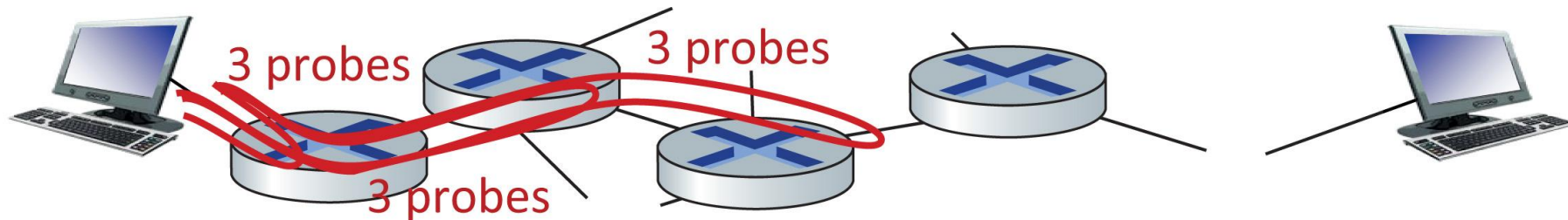


$La/R \sim 0$



$La/R \rightarrow 1$

- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply



traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

trans-oceanic link

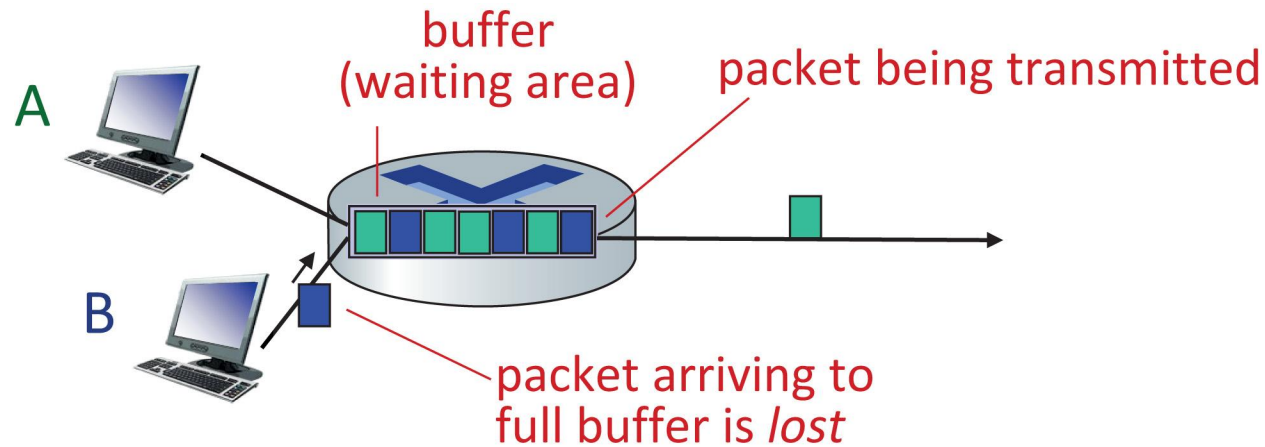
looks like delays
decrease! Why?

```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
```

* means no response (probe lost, router not replying)

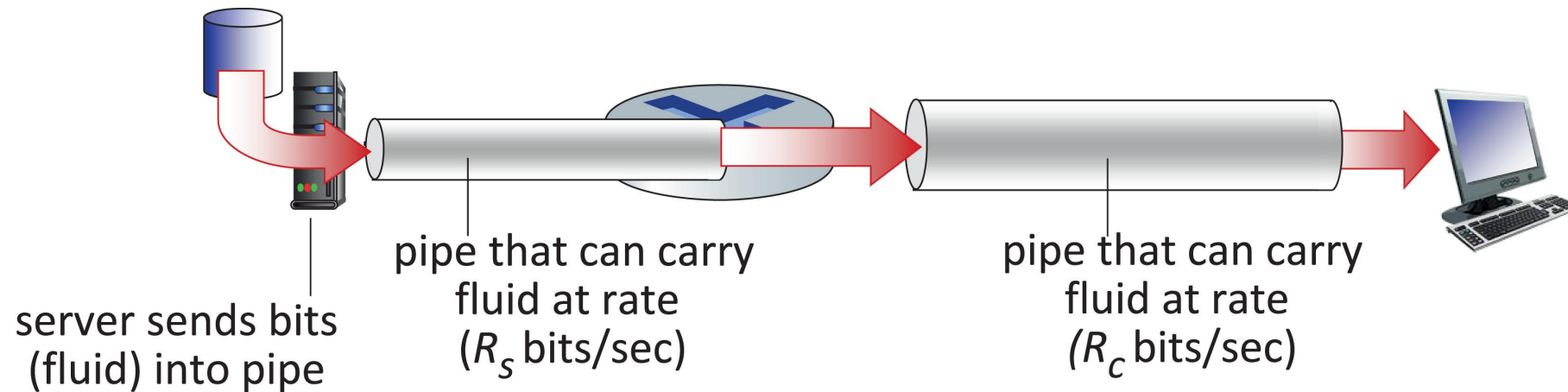
* Do some traceroutes from exotic countries at www.traceroute.org

- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all

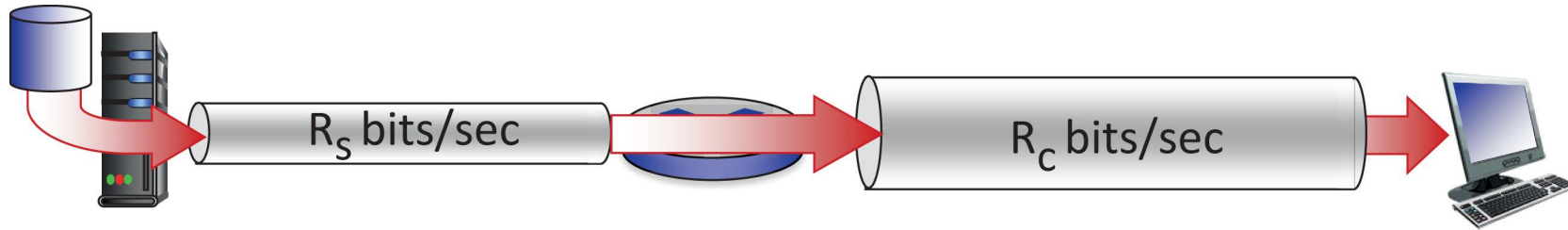


* Check out the Java applet for an interactive animation (on publisher's website) of queuing and loss

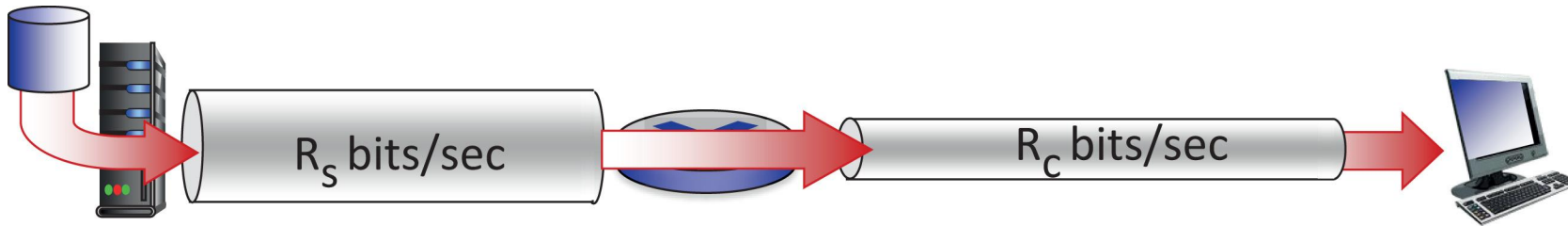
- **throughput:** rate (bits/time unit) at which bits are being sent from sender to receiver
 - **instantaneous:** rate at given point in time
 - **average:** rate over longer period of time



$R_s < R_c$ What is average end-end throughput?

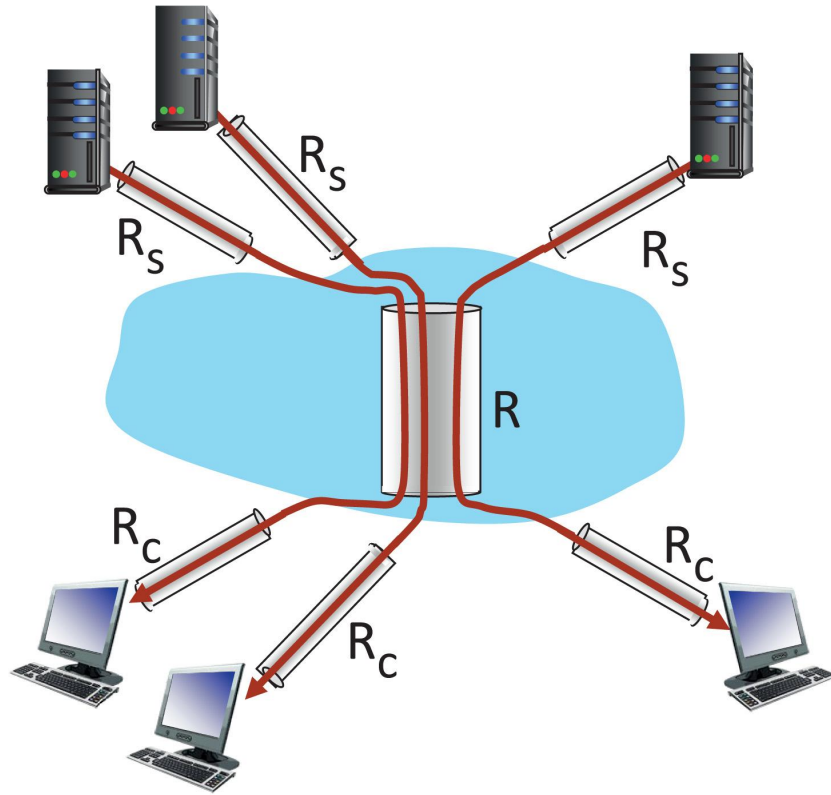


$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput



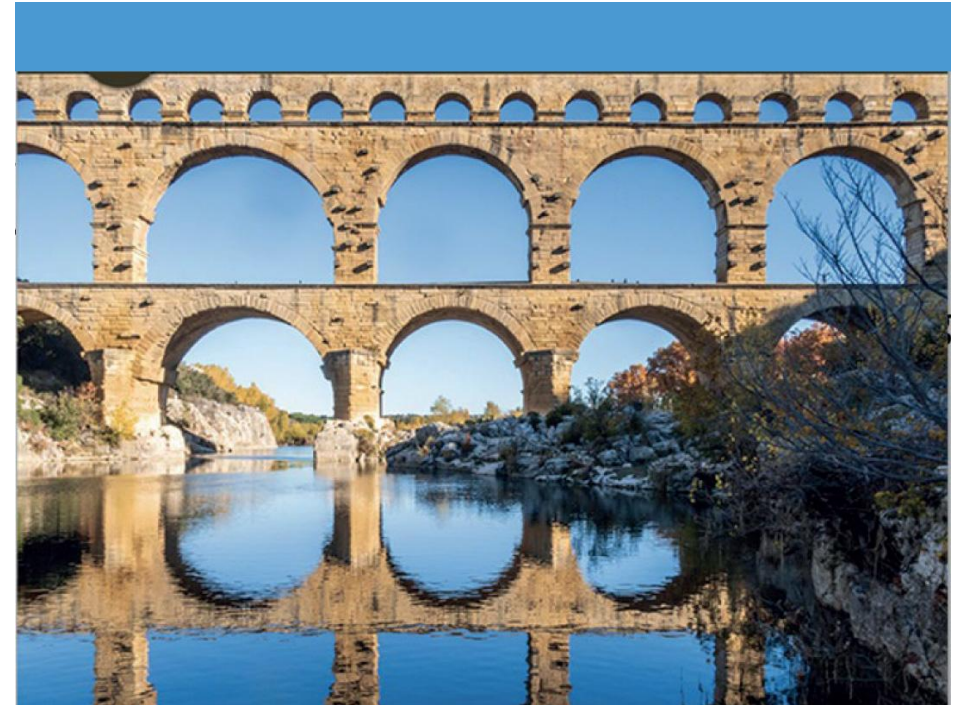
10 connections (fairly) share
backbone bottleneck link R bits/sec

- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

* Check out the online interactive exercises for more examples:

http://gaia.cs.umass.edu/kurose_ross/

- What is the Internet?
- What is a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Protocol layers, service models

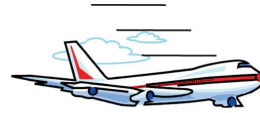


Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of **organizing** structure of network?

- and/or our **discussion** of networks?



————— *end-to-end transfer of person plus baggage* —————>

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

How would you **define/discuss** the *system* of airline travel?

- a series of steps, involving many services



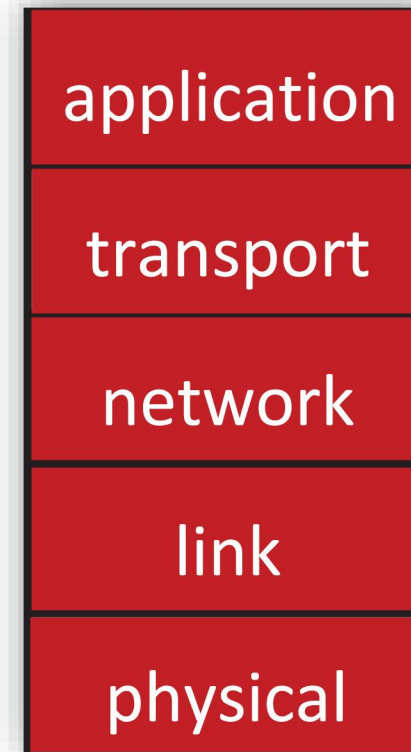
layers: each layer implements a service

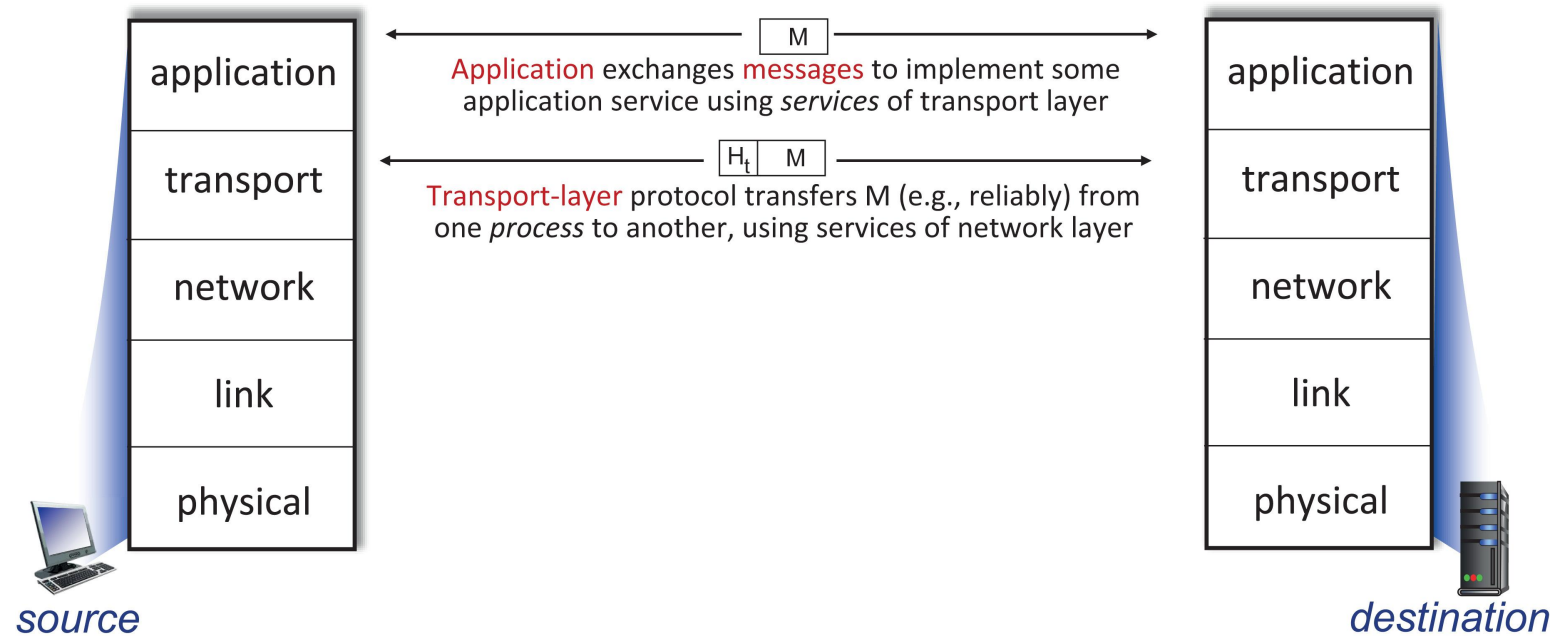
- via its own internal-layer actions
- relying on services provided by layer below

Approach to designing/discussing complex systems:

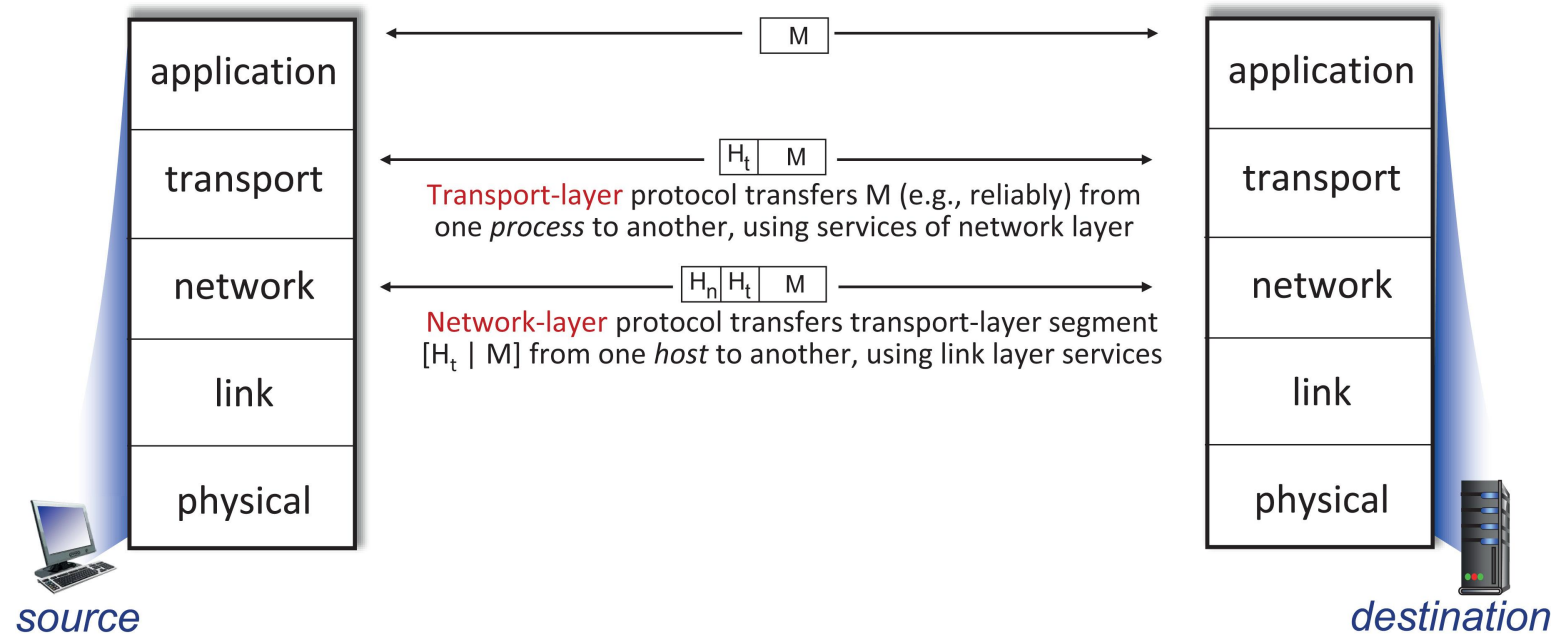
- explicit structure allows identification, relationship of system's pieces
 - layered **reference model** for discussion
- modularization eases maintenance, updating of system
 - change in layer's service **implementation**: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system

- **application:** supporting network applications
 - HTTP, IMAP, SMTP, DNS
- **transport:** process-process data transfer
 - TCP, UDP
- **network:** routing of datagrams from source to destination
 - IP, routing protocols
- **link:** data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- **physical:** bits “on the wire”

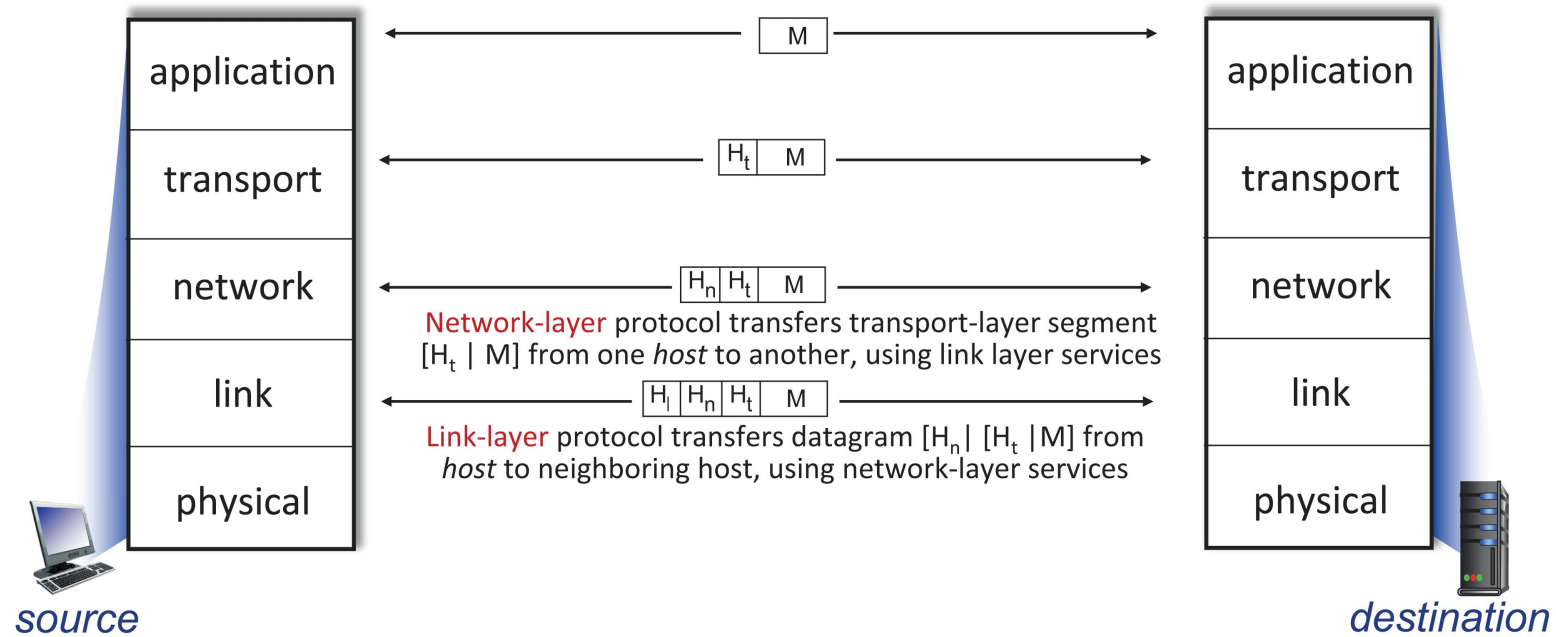




- transport-layer protocol **encapsulates** application-layer message, M, with **transport** layer-layer header H_t to create a transport-layer **segment**
- H_t used by transport layer protocol to implement its service

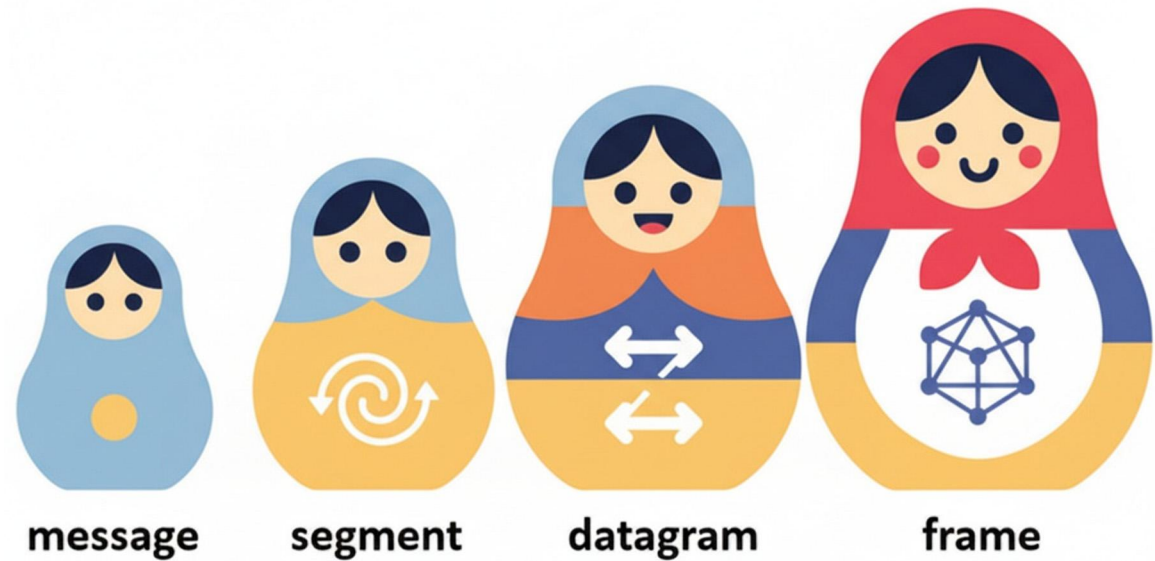


- network-layer protocol **encapsulates** transport-layer segment $[H_t | M]$ with network layer-layer header H_n to create a network-layer **datagram**
- H_n used by network layer protocol to implement its service



- link-layer protocol **encapsulates** network datagram $[H_n | [H_t | M]]$ with link-layer header H_l to create a link-layer **frame**

Matryoshka dolls (stacking dolls)



Credit: <https://dribbble.com/shots/7182188-Babushka-Boi>

